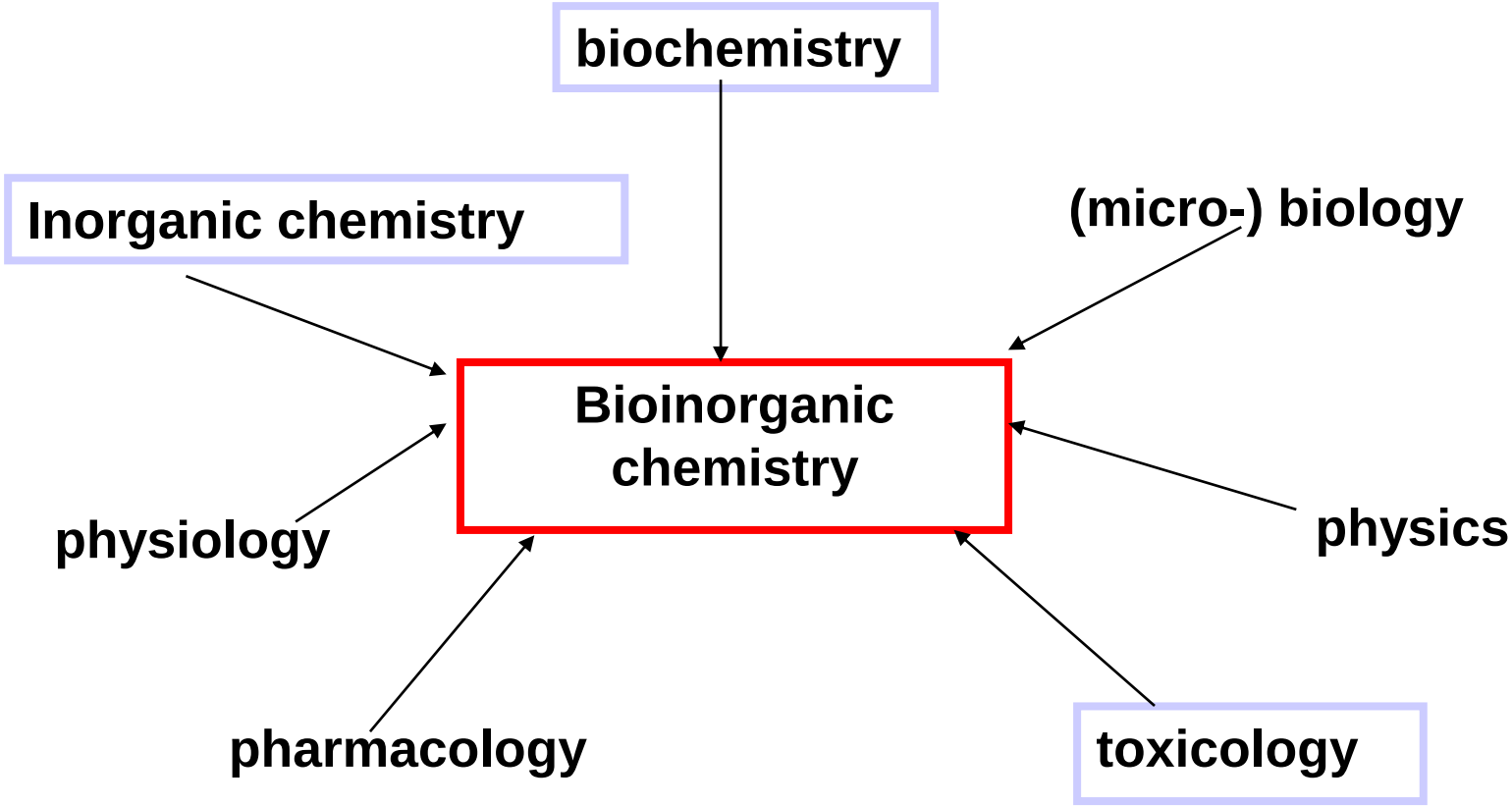
Bioinorganic Chemistry

What is bioinorganic chemistry?????

Study of inorganic elements in terms of their biological perspectives in order to understand the biological events



The Telegraph
calcutta, india
| Monday, October 01, 2007 |

An iron pill a day

Dr Gita Mathai (yourhealthgm@yahoo.co.in)

Some diseases are difficult to discover as they set in with no obvious date of onset. Anaemia is one such disease.

Anaemia causes the gradual onset of symptoms like tiredness, weakness, headache, light-headedness, shortness of breath, cold hands and feet, brittle nails, poor appetite and a sore tongue. The legs may be "restless", with frequent sleep-disturbing movements. Nutritive foods are ignored and there are bizarre cravings for non-food items like ice, uncooked rice, chalk, coal, paint, and powder.

India has the largest number of people with iron deficiency anaemia. About 60-70 per cent of the population in all age groups is affected. Their haemoglobin is 10g/dL or less. (Normal levels of haemoglobin range between 11 and 15g/dL.)

Iron is required for the production of haemoglobin, the oxygen-carrying component of blood found in red blood corpuscles (RBCs). A deficiency causes a change in the number and size of the RBCs.

Men lose 1 milligram of iron a day. Women lose 1.5 to 2 milligram of iron a day.

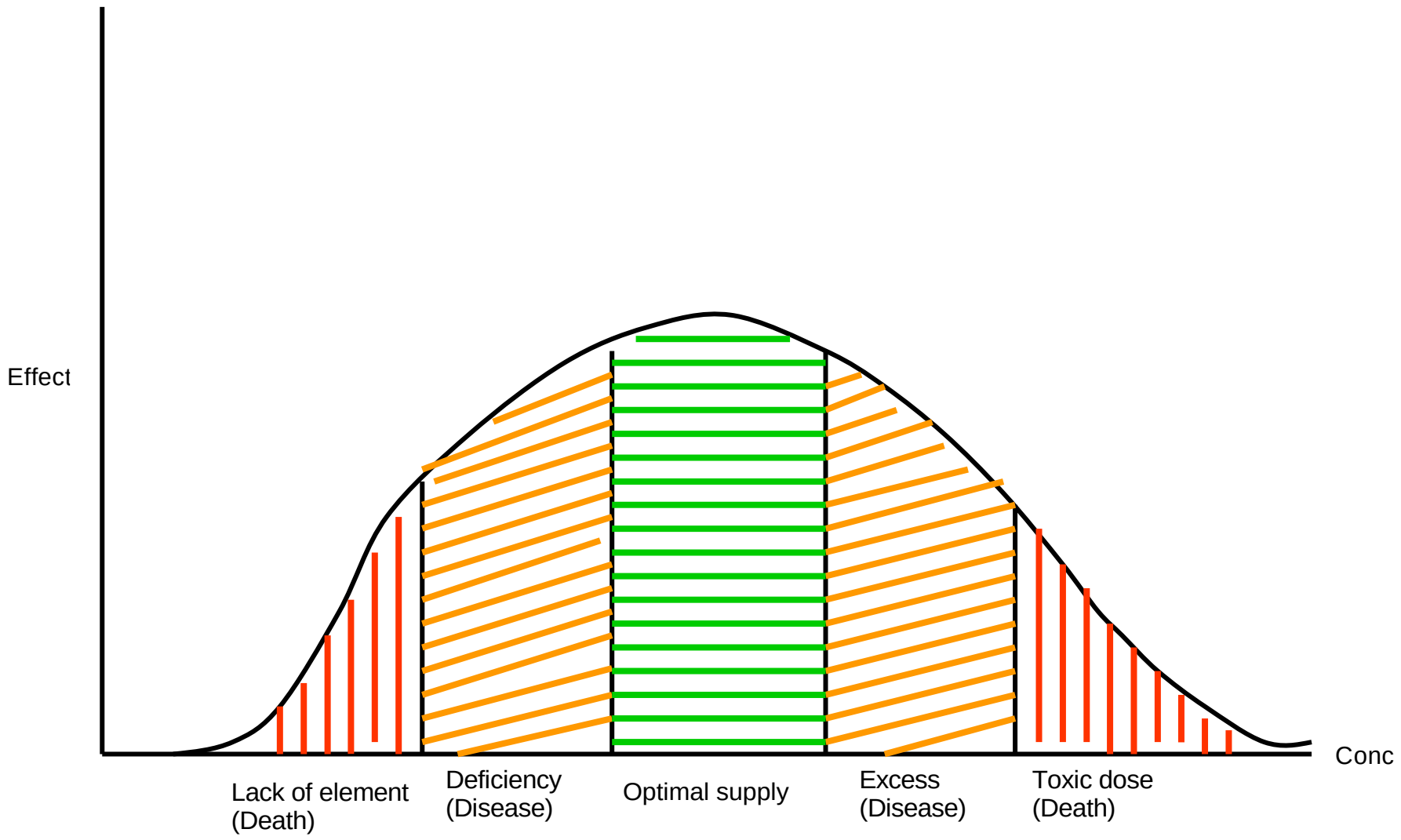
Women are more likely to be affected by iron deficiency anaemia than men.

Adolescents, are even more prone to anaemia.

The requirement of iron is 10-15 mg a day.

When advised, iron pills should be taken once in a day. Take it with lime or orange juice, as an acidic medium increases absorption. Do not combine it with calcium or zinc supplements as this interferes with absorption.

Concentration and physiological effect



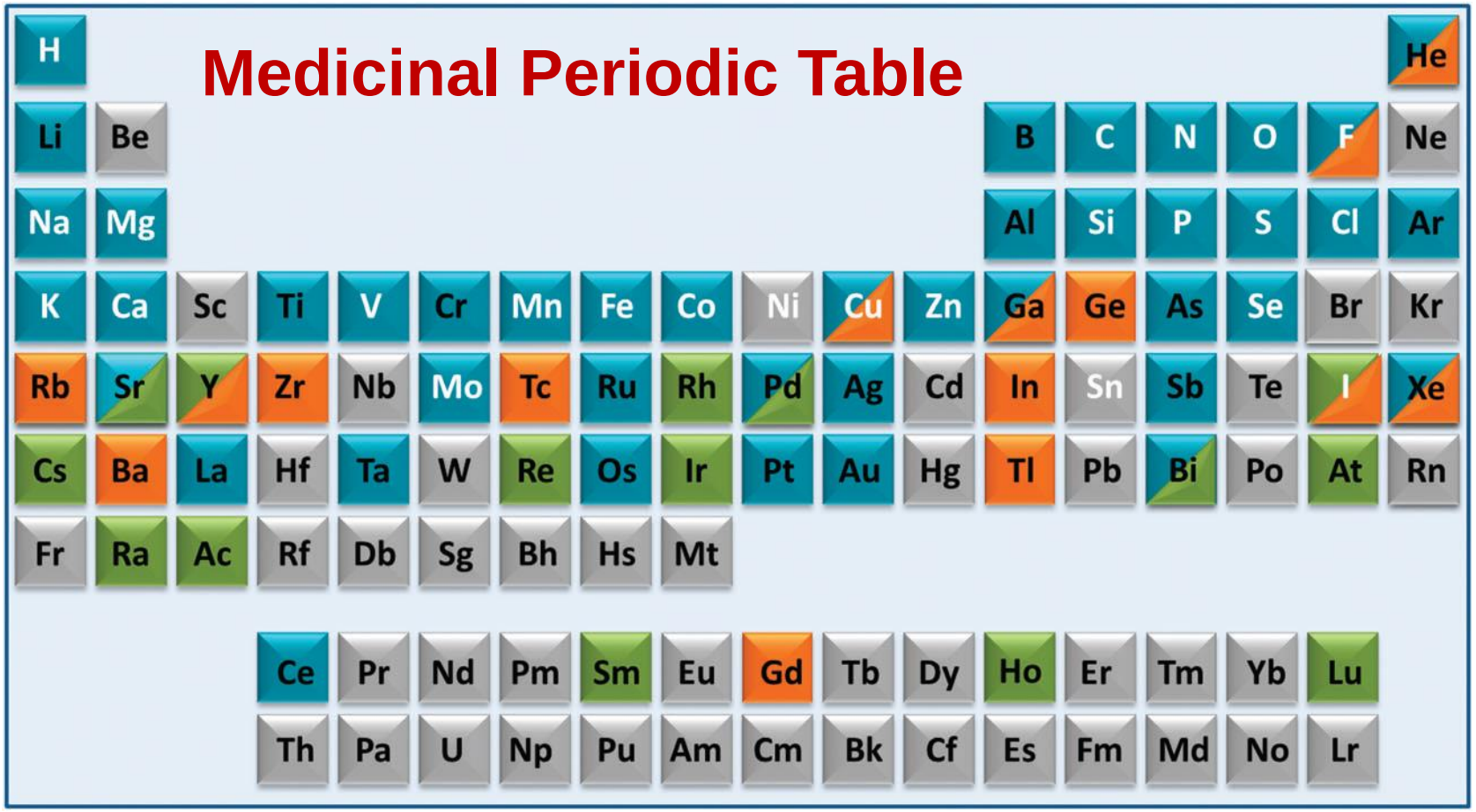


Fig. 1 A medical periodic table: essential elements for man (**symbols in white font**); medical radioisotopes (**green fill**); elements currently used in therapy (**blue fill**) or diagnosis (**orange fill**). The entries (**limited to 2 fill colours**, illustrative and not comprehensive) are mainly restricted to elements/compounds which are clinically approved or on current clinical trials (e.g. as listed on http://www.clinicaltrials.gov/). Some entries for implants are included (e.g. Ti, Ta).

Elements in biology

H																	He
Li	Be											B	C	N	O	F	Ne
Na	Mg											Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Fr	Ra	Ac															

Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr

Metals — Essential elements for humans (daily requirement: 25 mg)

Non metals — Presumably essential elements

Non metals other than C, H, N, O, S in biology

- Si^{IV}:
- P^{V}:
- Se^{-II}:
- F^{-}:
  ani

Metal ions in biology

- Na^{+},K^{+}:
- Mg^{2+}:
- Ca^{2+}: $_3$
- $V^{IV/V}$, $Mo^{IV/VI}$, $W^{IV/VI}$, $Mn^{II/III/IV}$, $Fe^{II/III}$, $Ni^{I/II/III}$, $Cu^{I/II}$:
- Fe and Cu:
- Fe^{II}, Fe^{III}:
- Co:
- Zn^{2+}:

Suggested daily intake							
V	<0.1 mg	Cr	0.06	Co	0.3	Mo	0.2
Ni	0.4	Mn	3	Cu	3		
Zn	12	Fe	15				

Metal ions in biology

- Main group elements: K, Na, Ca, Mg
- Transition metals: Fe, Cu, Mn, V, Cr, Ni, Zn, Mo etc.
- Roles:
- Electron transfer
- Bind small molecules (e.g. O_2) and take part in their transport
- Templates for syntheses of biomolecule

Non-metal ions in biology

$O_2/O_2^{\bullet-}/O_2^{2-}$

oxyMb: $Fe^{II}(O_2)^0$
oxyHb: $Fe^{III}(O_2^{\bullet-})$

$$O{=}O \underset{-e^-}{\overset{+e^-}{\rightleftharpoons}} O{=}O^{\bullet-} \underset{-e^-}{\overset{+e^-}{\rightleftharpoons}} {}^-O{-}O^-$$

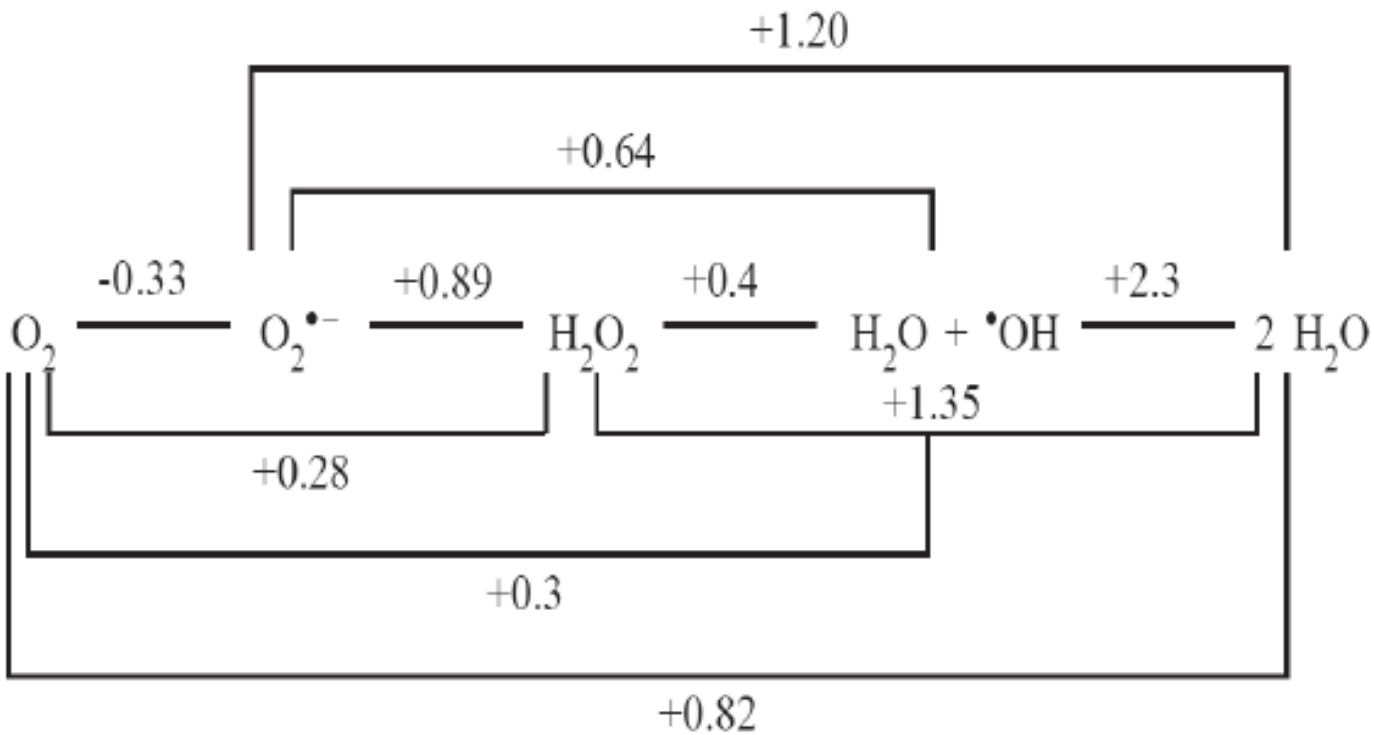
dioxygen superoxide peroxide

Non-metal ions in biology

Variations on O_2

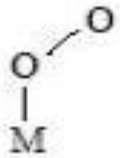
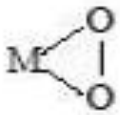
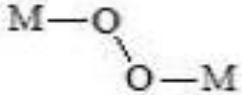
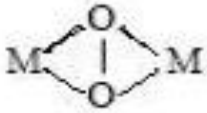
	triplet	singlet	superoxide	peroxide	oxide	oxyl
	$^3O_2$	$^1O_2(^1\Sigma)$, $^1O_2(^1\Delta)$	$^2O_2^{\bullet-}$	$^1O_2^{2-}$	O^{2-} +	$O^{\bullet-}$
+ H^+ →			$HO_2^{\bullet}$	HO_2^-	OH^-	$^{\bullet}OH$
+ 2 H^+ →				H_2O_2	H_2O	

Non-metal ions in biology



(redox potentials in V vs. normal hydrogen electrode, pH 7)

Metal coordination of O_2^{n-}

structure:	coordination mode of O_2^{n-}		example:
	η^1	end-on	$[(CN)_5Co(O_2)]^{3-}$
	η^2	side-on $(HBPz^*_3)Co(O_2)$:	$(Ph_3P)_2(CO)(Cl)Ir(O_2)$ Egan et al.
	$\mu, \eta^1 : \eta^1$	end-on bridging	$[(H_3N)_5Co(O_2)Co(NH_3)_5]^{5+}$
	$\mu, \eta^2 : \eta^2$	side-on bridging	$[(Cl_3O_2U)_2(O_2)]^{4-}$ $[(HBPz^*_3)Cu]_2(O_2)$: Kitajima, Moro-oka

Biological Functions of O2

Biological Functions of O_2

- uptake, transport and storage (reversible)
- energetic use:

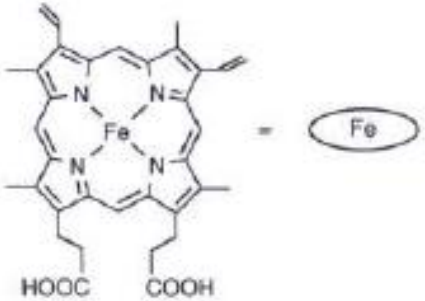
- monooxygenation $O_2 + X \rightarrow XO + H_2O$

- dioxygenation $O_2 + 2\,X \rightarrow 2\,XO$

- oxidation $O_2 + X \xrightarrow{2e^-} H_2O_2 + [X^{2+}]$

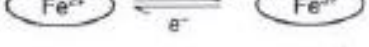
$O_2 + X \xrightarrow{4e^-} H_2O + [X^{4+}]$

heme
(Fe-protoporphyrin IX)

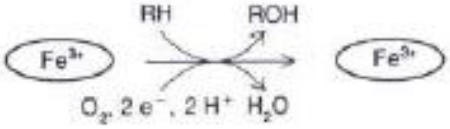


hemoglobin,
myoglobin
(Section 5.2)

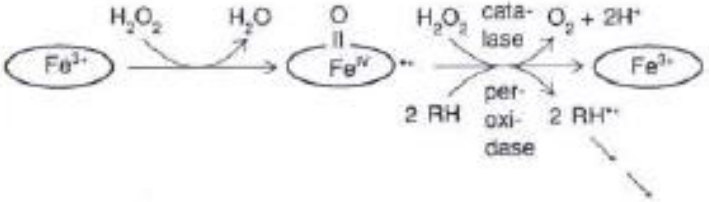
cytochromes
(Section 6.1)



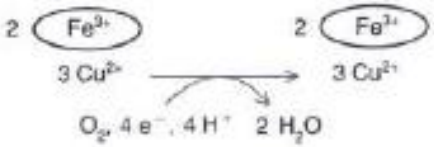
cytochrome P-450
(Section 6.2)



catalase and
peroxidase
(Section 6.3)



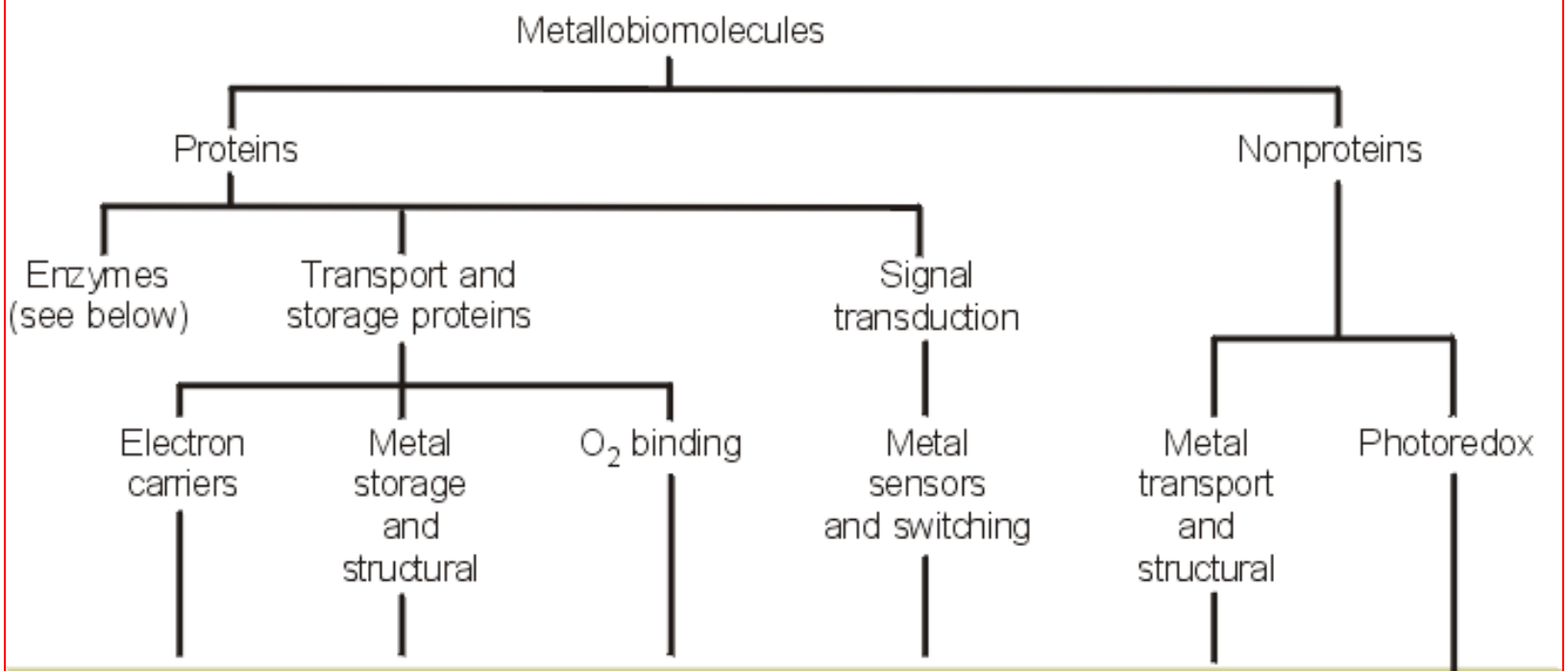
cytochrome *c*
oxidase
(Section 10.4)



(5.8)

Metallobiomolecules

Table 19.2 The classification of some biomolecules containing metal ions



Metalloenzymes

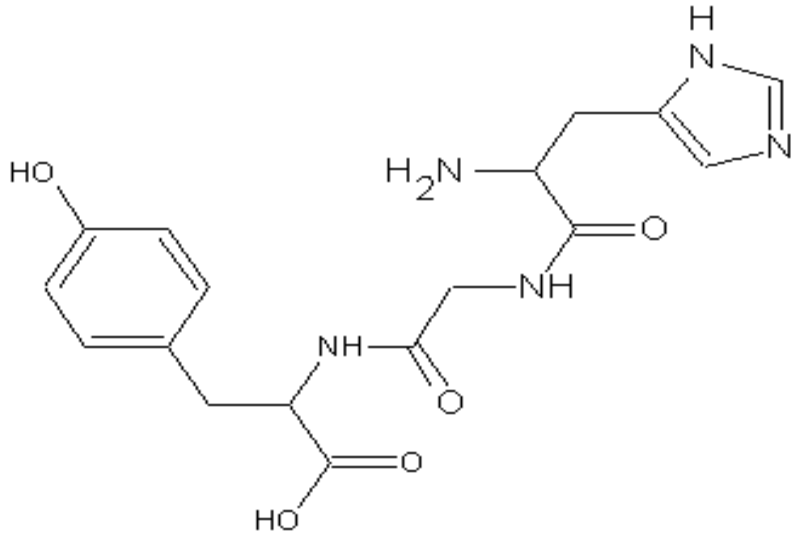
Enzymes

	Hydrolases	Oxido-reductases	Isomerases and synthases
Fe:		Oxygenases Hydrogenases	Aconitase Endo III
Fe, Mo:		Nitrogenases	
Fe, Mo, Cu:		Oxidases Reductases Hydroxylases	
Mo, Cu, Zn:		Superoxide dismutases	
Co:			Vitamin B_{12} coenzymes
Zn:	Carboxypeptidase		
Zn, Cu, Mg:	Phosphatases		
Mg, Mn:	Aminopeptidases		

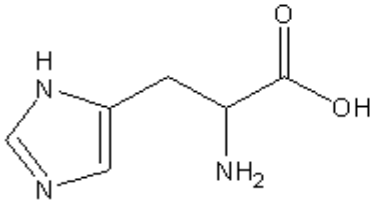
Augmented version of J.A. Ibers and R.H. Holm, *Science* **209**, 223 (1980)

Metal Binding Groups

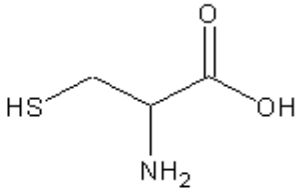
functional groups like
-COOH, -OH, -SH, NH_2
$CONH_2$ in peptide bond



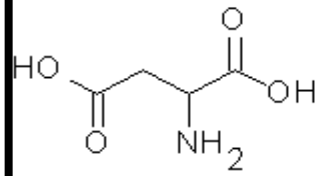
Tyrosine (Tyr)Histidine (His)



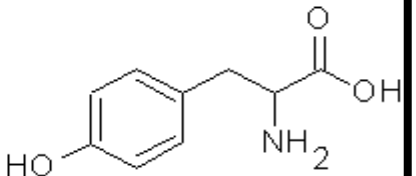
Histidine (His)



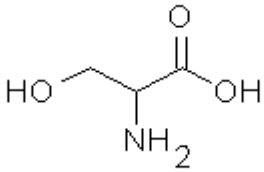
Cysteine (Cys)



Aspartic Acid (Asp)

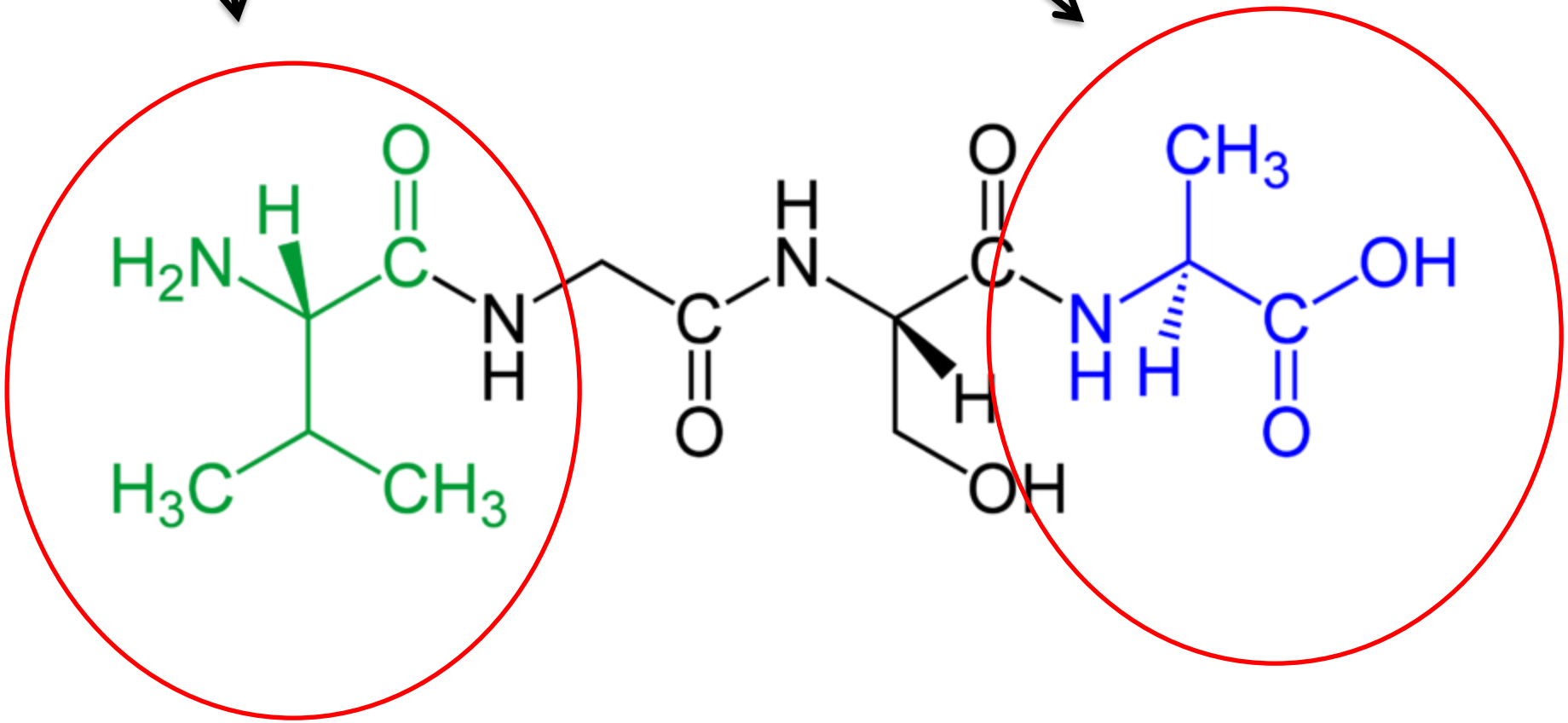


Tyrosine (Tyr)

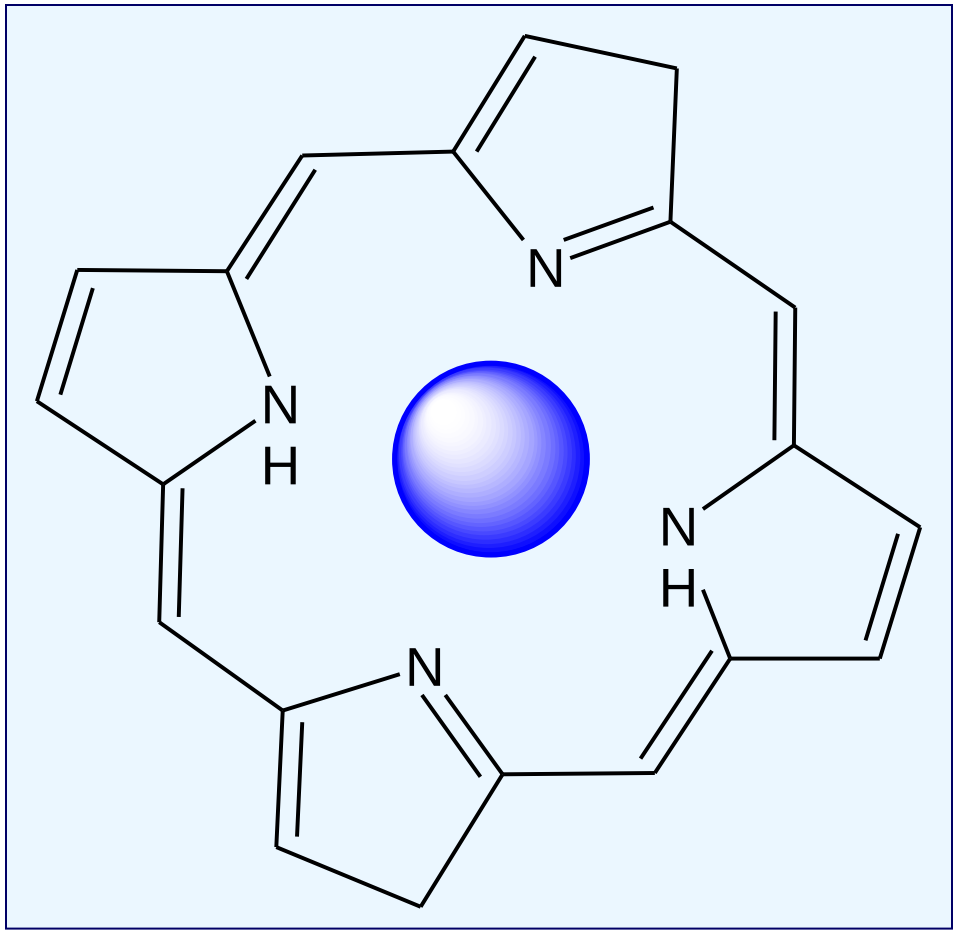


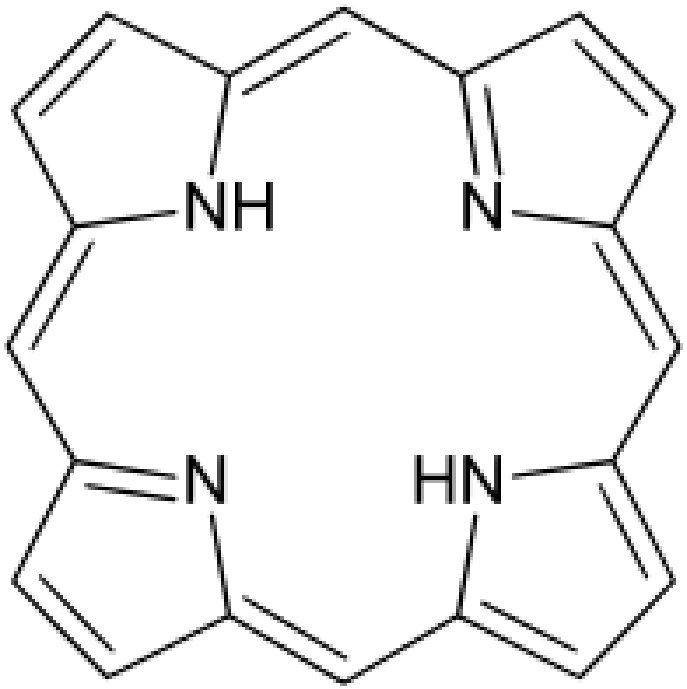
Serine (Ser)

N – terminal and C-terminal group



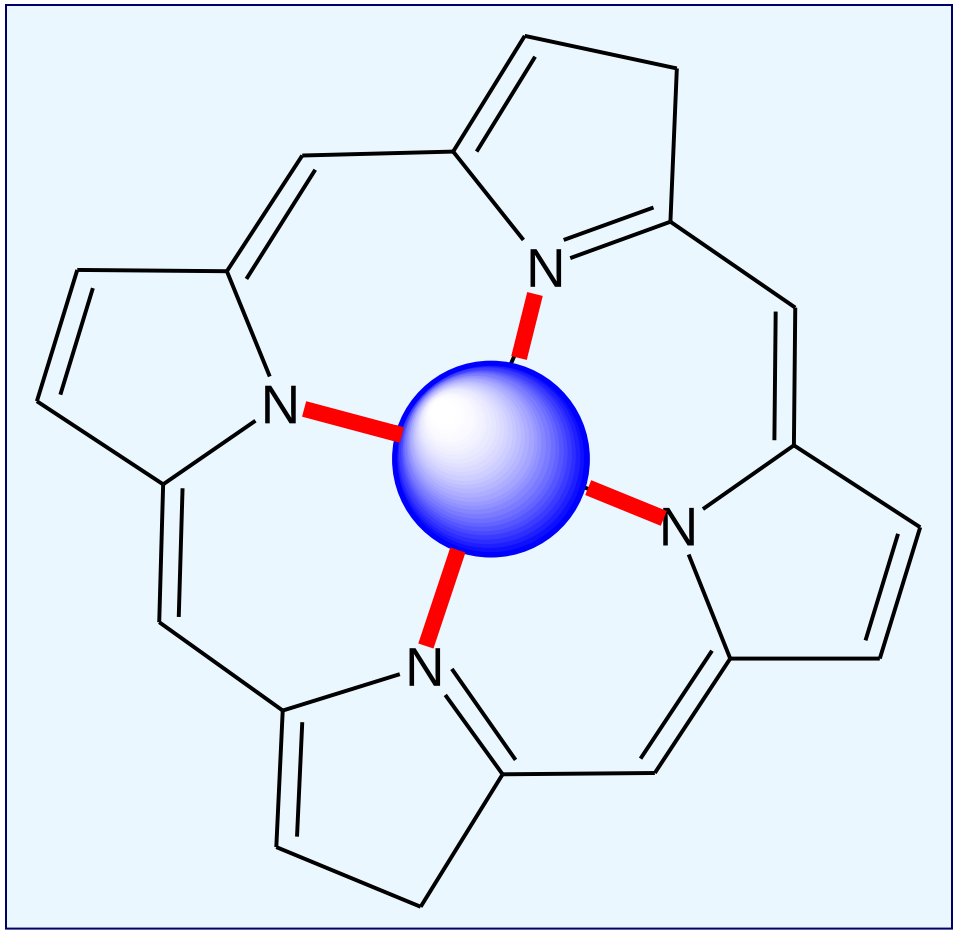
PORPHYRIN: A STRONG METAL BINDING GROUP





Final structure of porphirin

PORPHYRIN: A STRONG METAL BINDING GROUP



Fairly rigid

M-N bond length:

196 pm - Ni(II)
210 pm - Fe(II)

Biological oxygen carrier

- **The problem: Oxygen is required by cells throughout organisms for respiration**
- **But, diffusion of oxygen through tissues is ineffective**
- **Also, oxygen has low solubility in aqueous solutions**
- **To solve this, we need a carrier molecule that can transport oxygen from the lungs to tissues where the oxygen is needed**

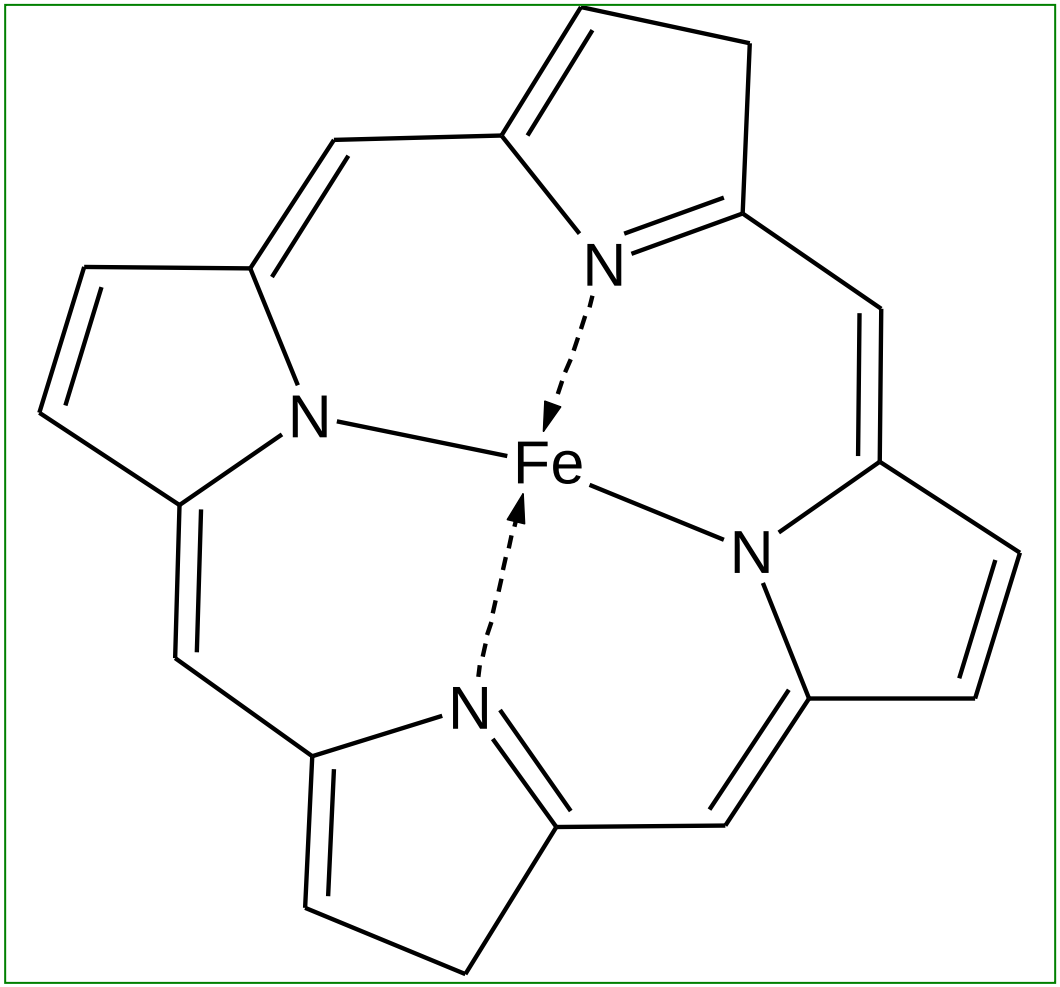
Biological oxygen carrier

- Should bind oxygen reversibly
- Should efficiently pick up oxygen in the lungs (high capacity and fast binding)
- Should efficiently release oxygen in tissues where it is needed
- Should cause no side reactions
- Organic molecules do not suit for reversible oxygen binding
- Metals could be better candidates. But reaction of metal complexes with oxygen can be irreversible
- One needs to worry about the production of dangerous free radicals as well

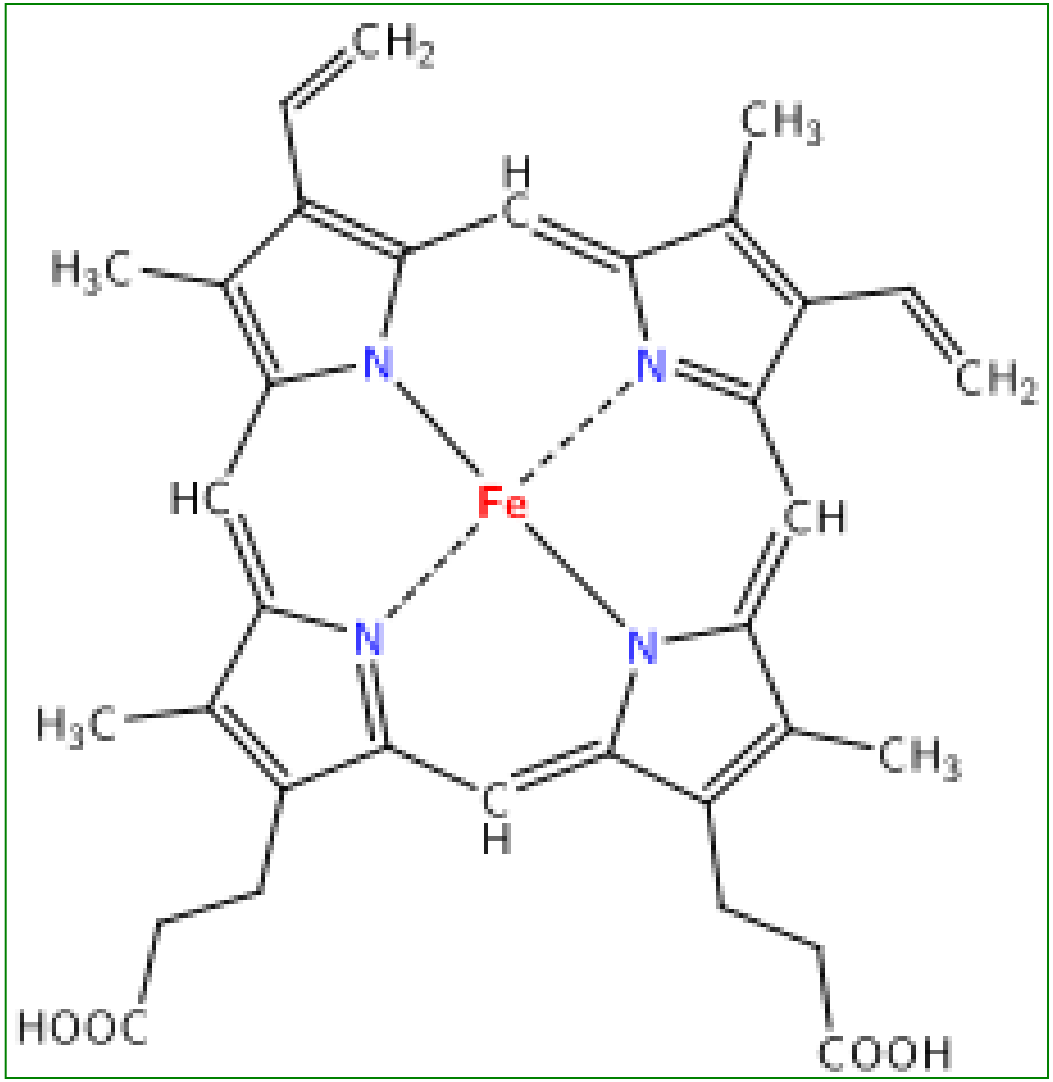
The solution: Hemoglobin

Porphyrin bound Fe: Heme

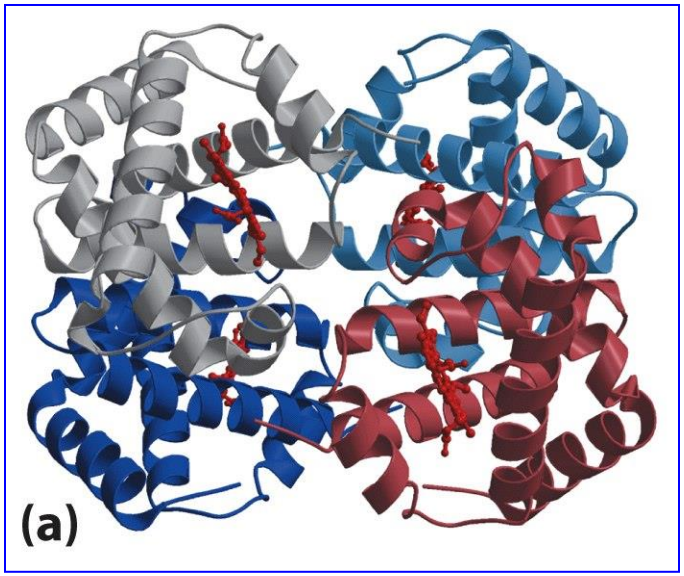
Heme Protiens: Hemoglobin



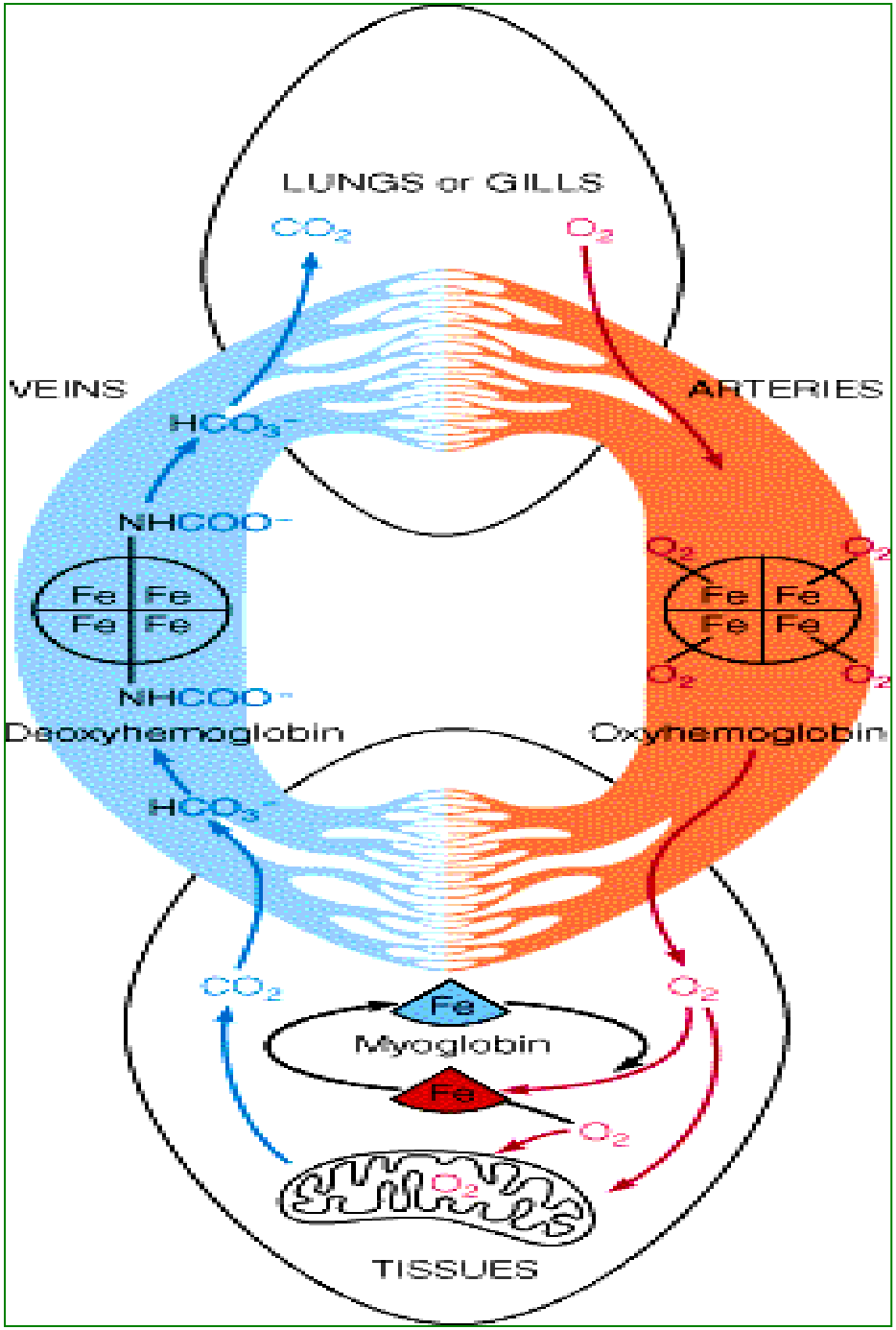
Porphyrin bound Fe: Heme **Heme Protiens: Hemoglobin**



Hemoglobin & O2 Binding



- Hemoglobin has four heme (iron-porphyrin) groups that bind oxygen.
- The protein surrounding the hemes plays important roles.
- Prevents side-reactions like irreversible heme oxidation
- Enhances selectivity for O_2 binding (relative to CO for example)
- Tetramer allows cooperativity making hemoglobin efficient.



Hemoglobin

⇓

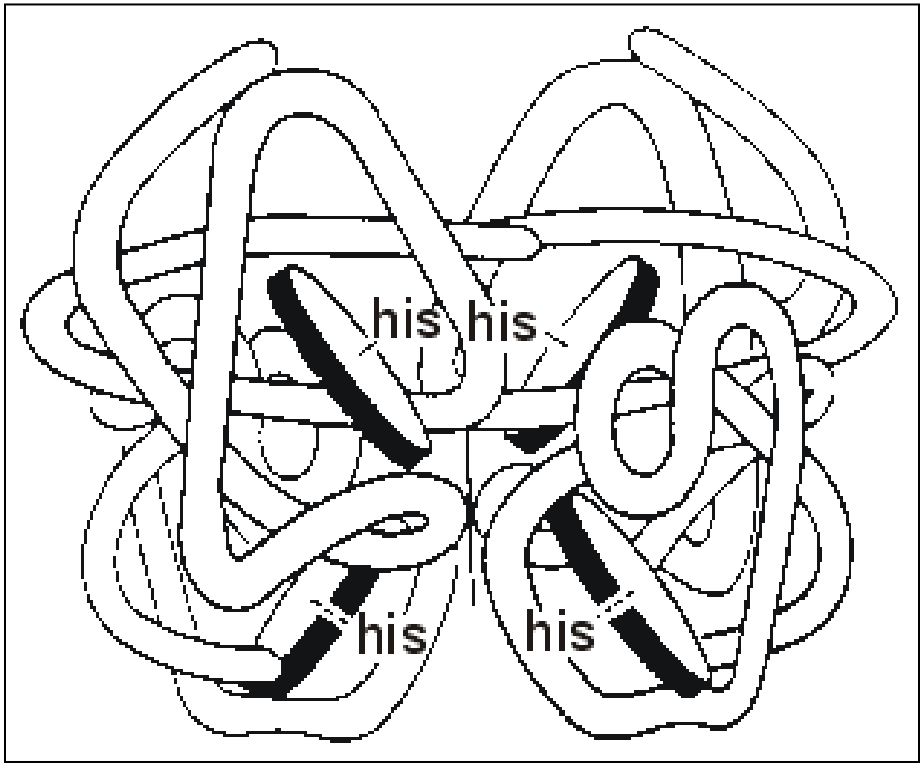
Transport O_2 from lung to tissues

Myoglobin

⇓

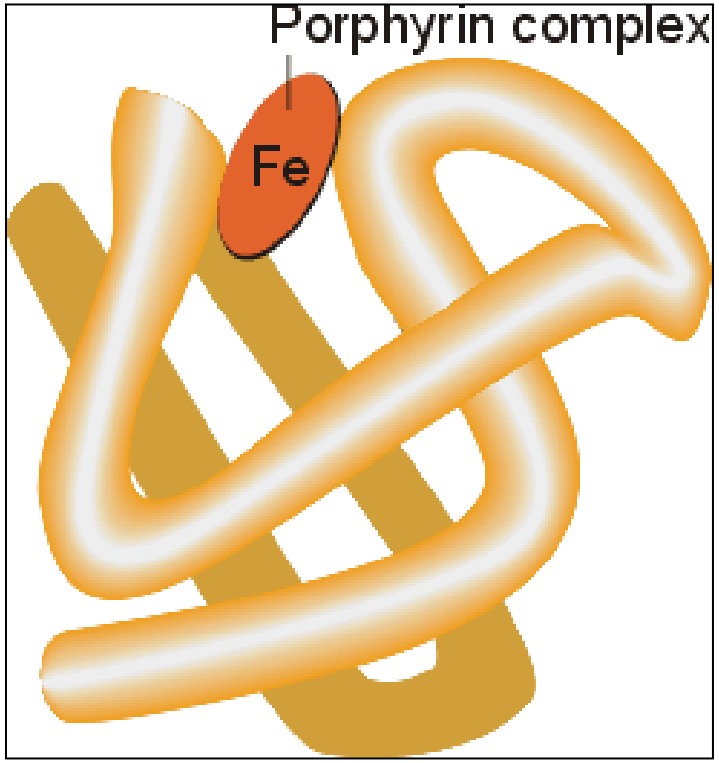
O_2 Storage

HEMOGLOBIN



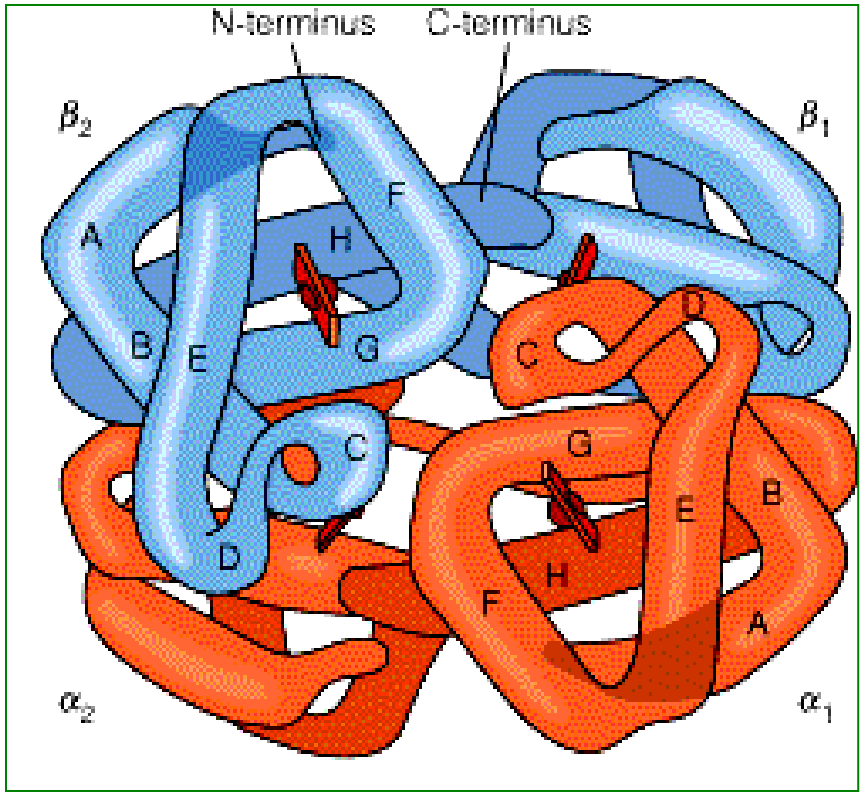
M.W. 64500
Tetrameric protein
β- 146 amino acids
α- 141 amino acids

MYOGLOBIN



M.W.: 17,000
153 amino acid residue
single heme group

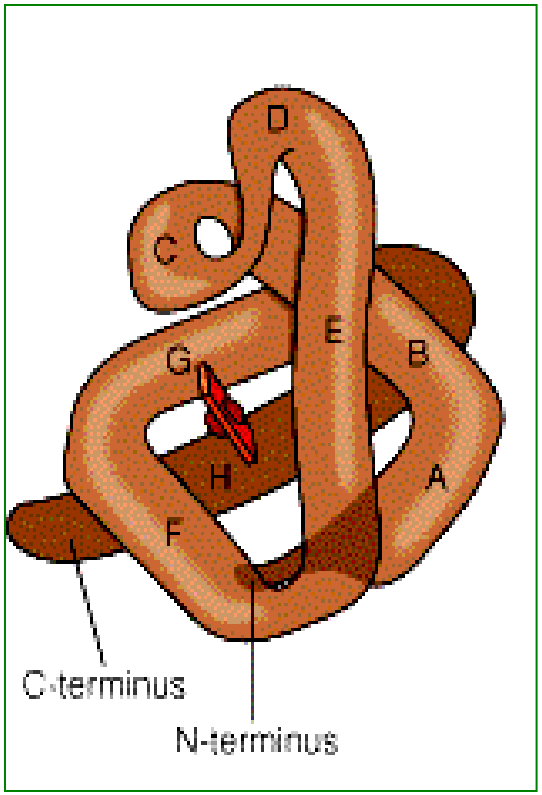
HEMOGLOBIN



α – helix

β – helix

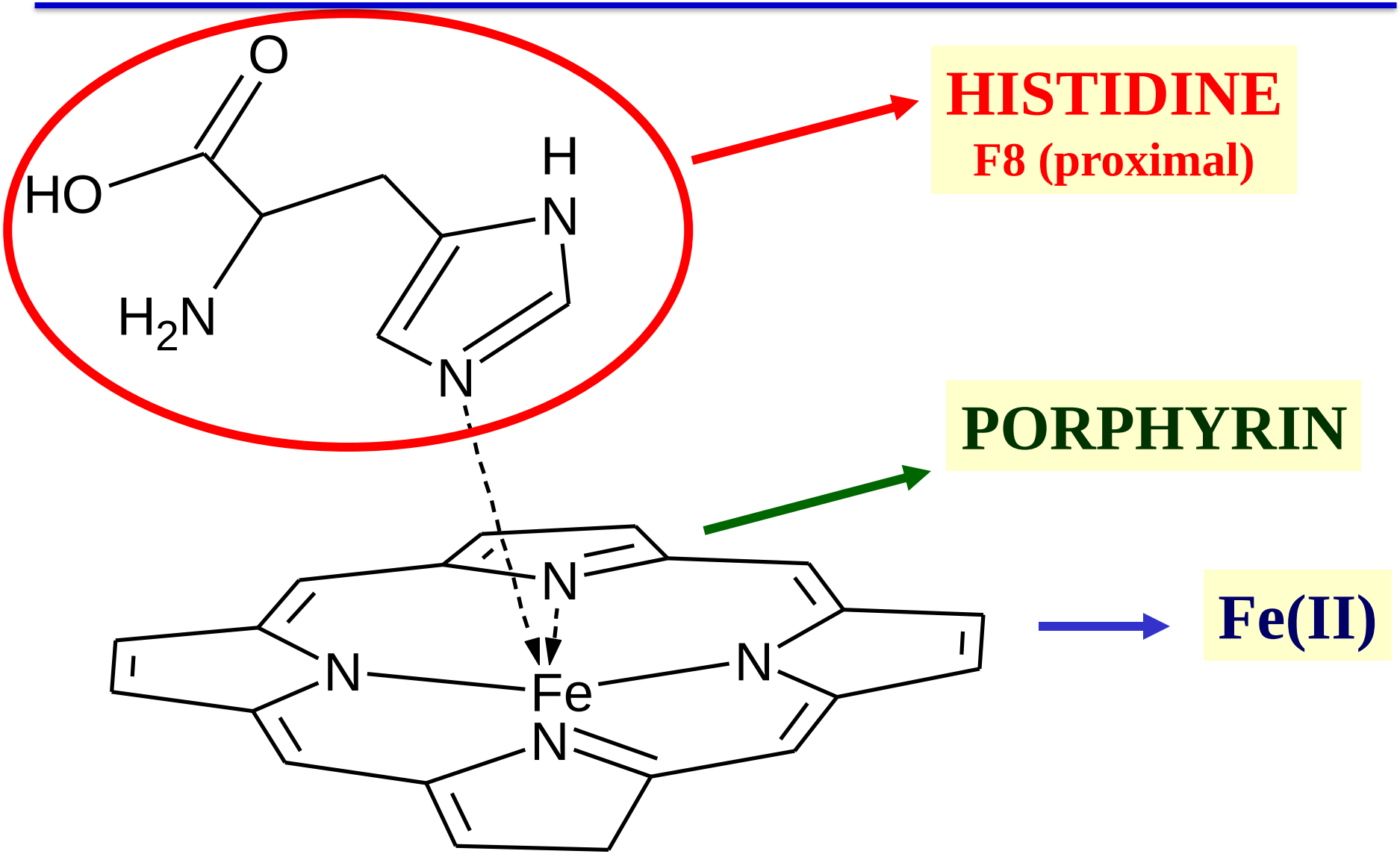
MYOGLOBIN



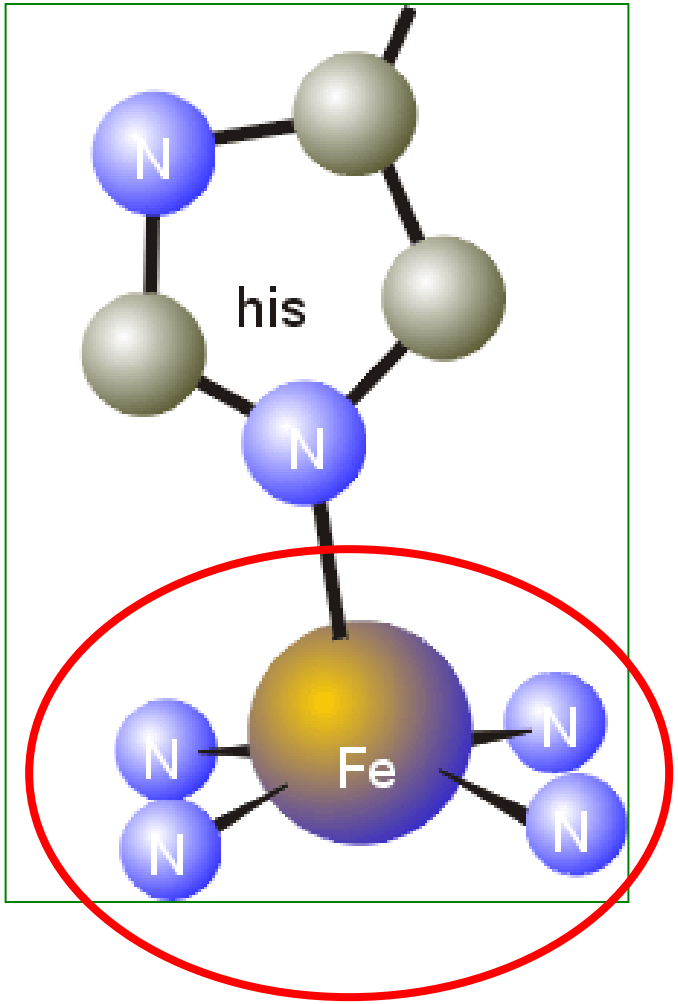
Fe(II) d^6

5 coordinate – high spin

Before Oxygen BINDS



Before Oxygen BINDS

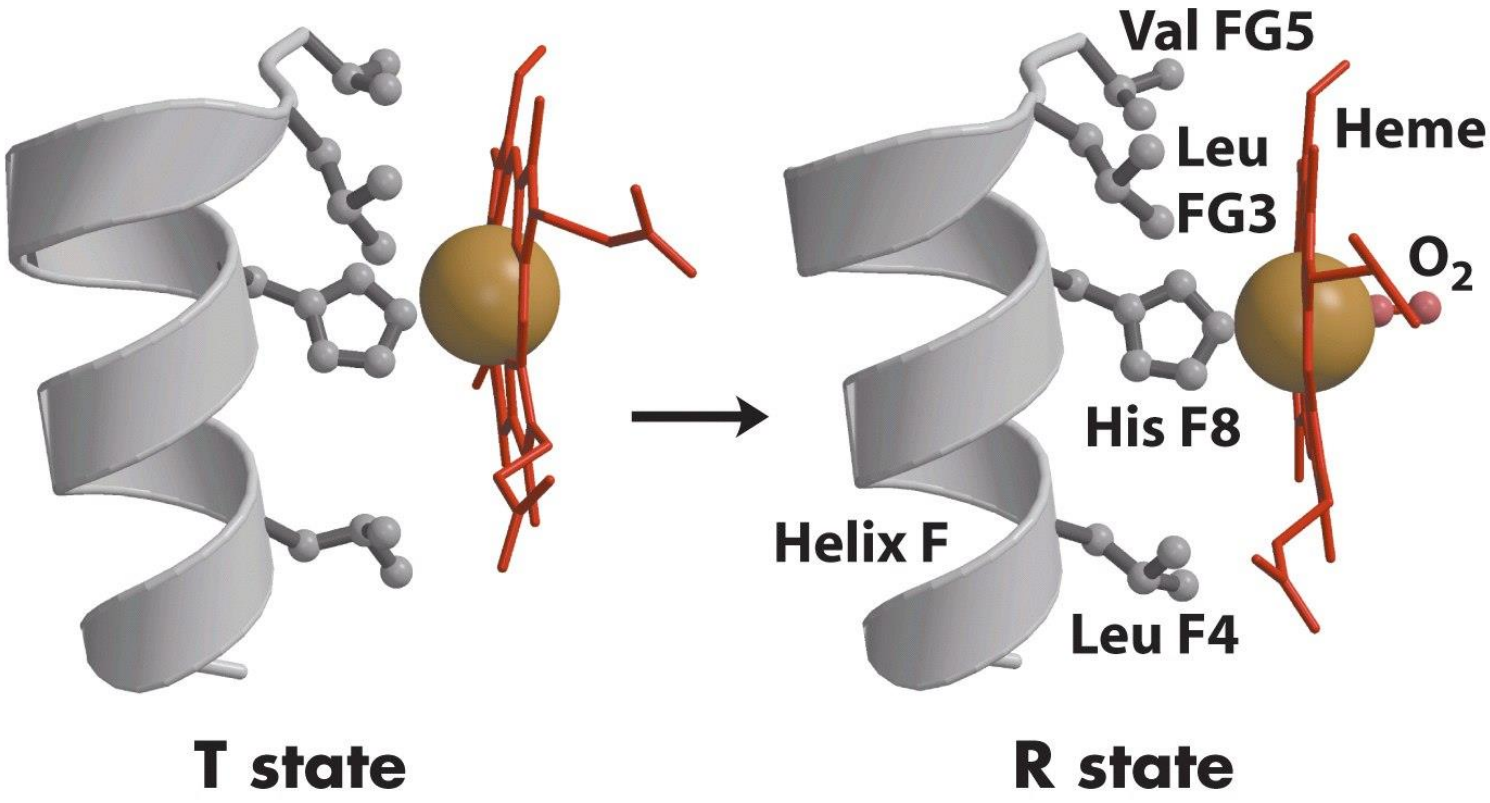


O2 Binding & Hemoglobin dynamics

- A Quaternary Structure Change
- When deoxy-Hb crystals are exposed to oxygen, they shatter! Evidence of a structural change!
- One alpha-beta pair moves relative to the other by 15 degrees upon oxygen binding
- This massive change is induced by movement of Fe by 0.039 nm when oxygen binds.
- Two of the heme move 100 pm towards each other.
- Other two separate by about 700 pm.

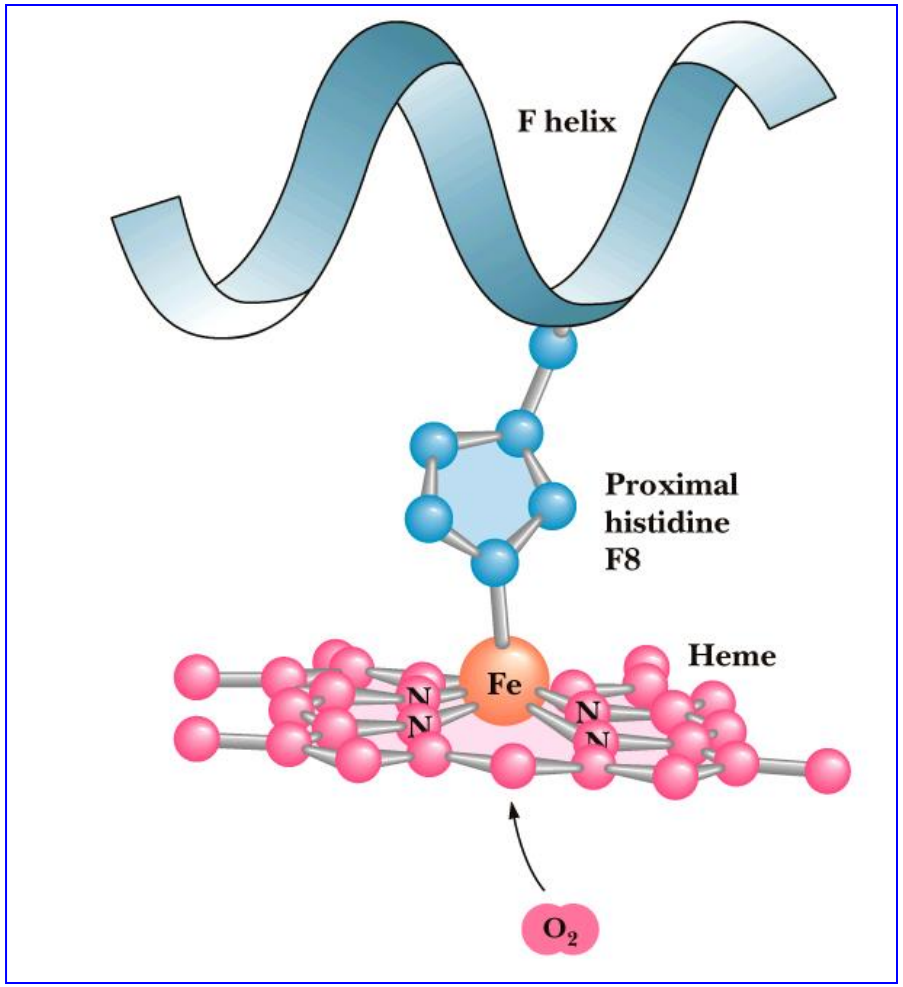
O2 Binding & Hemoglobin dynamics

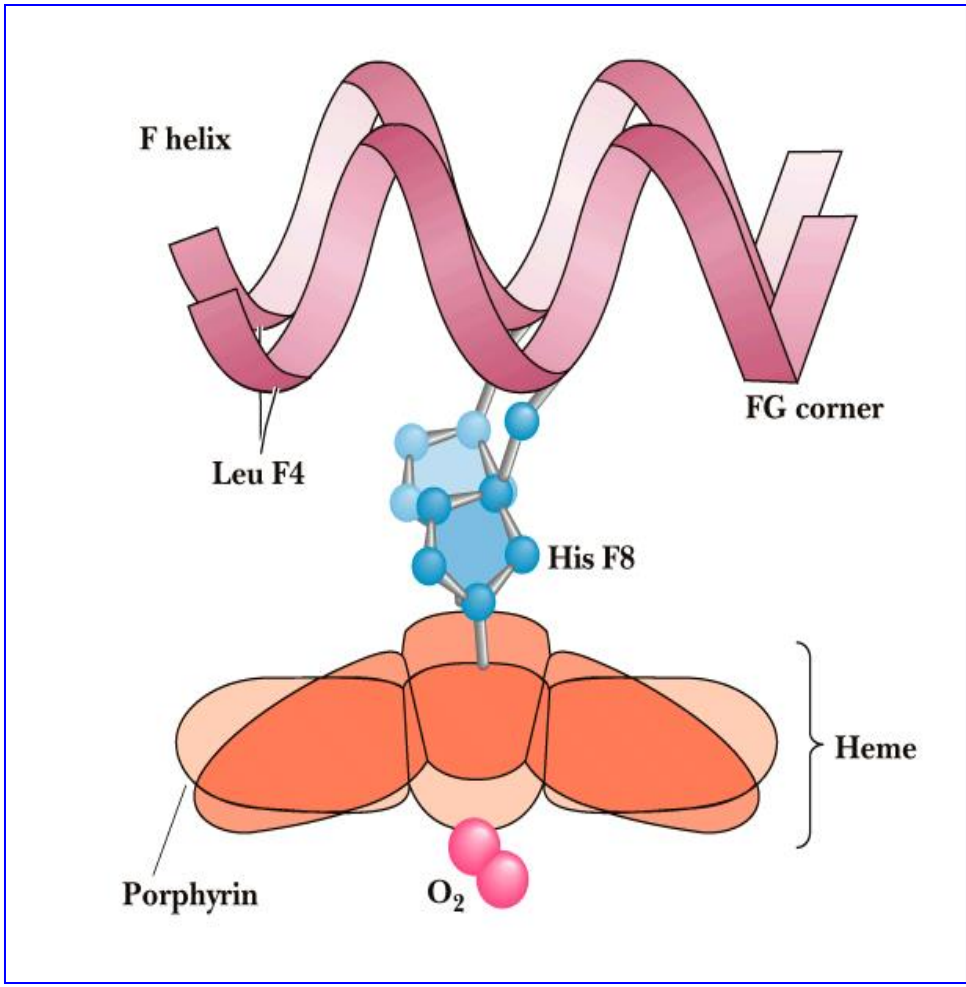
Oxygen binding triggers local conformational changes:



The shift in the iron tugs on the His, which moves helix F

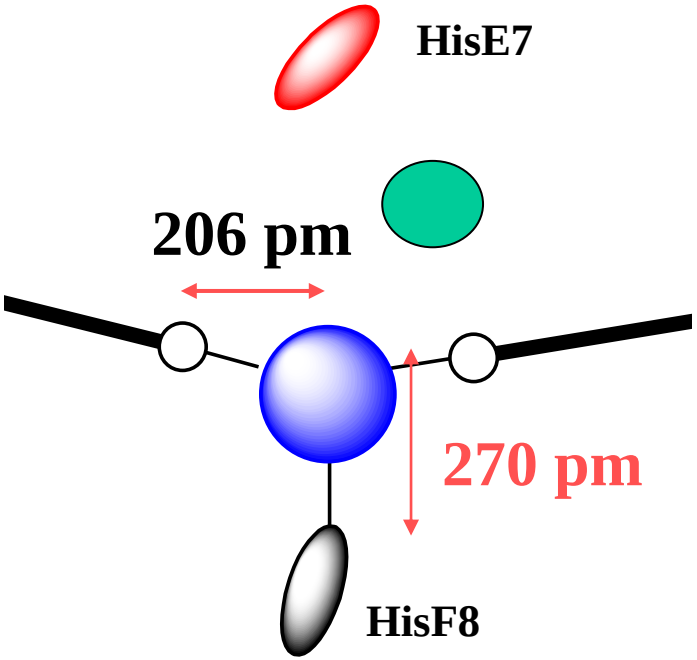
O2 Binding & Hemoglobin dynamics



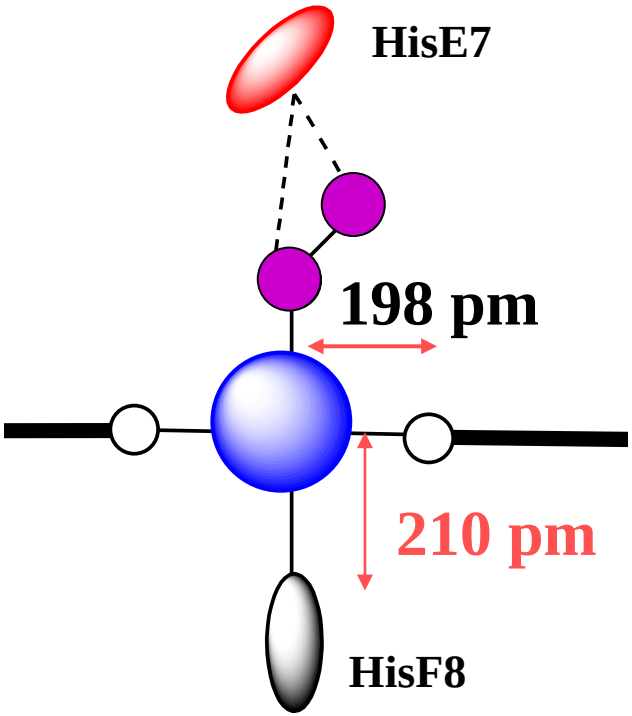


O2 Binding & Hemoglobin dynamics

T- deoxy state



R- oxy state



T-state:

Fe(II) d^6 $t_{2g}{}^4 eg^2$ high spin

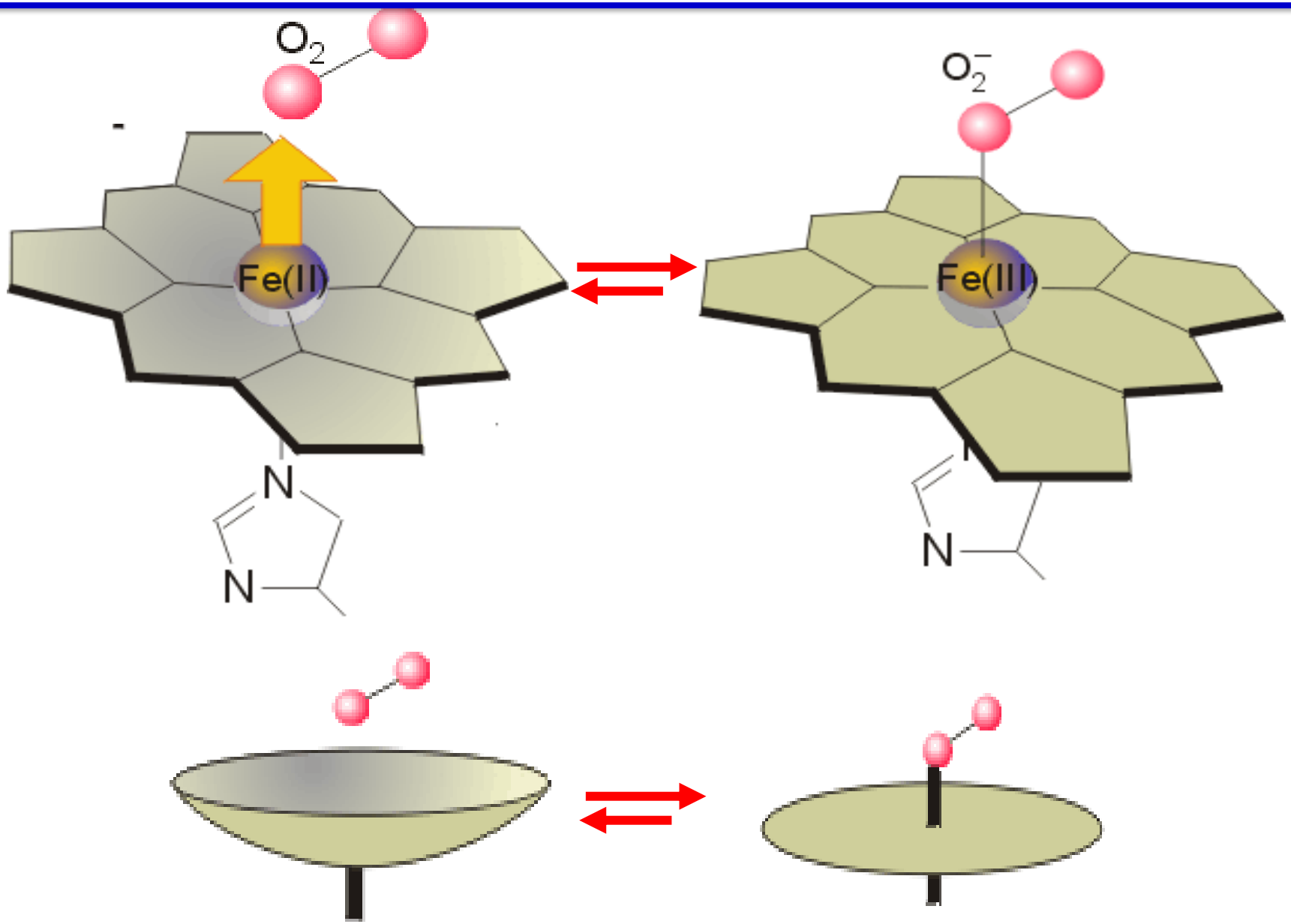
Radius: 92 pm

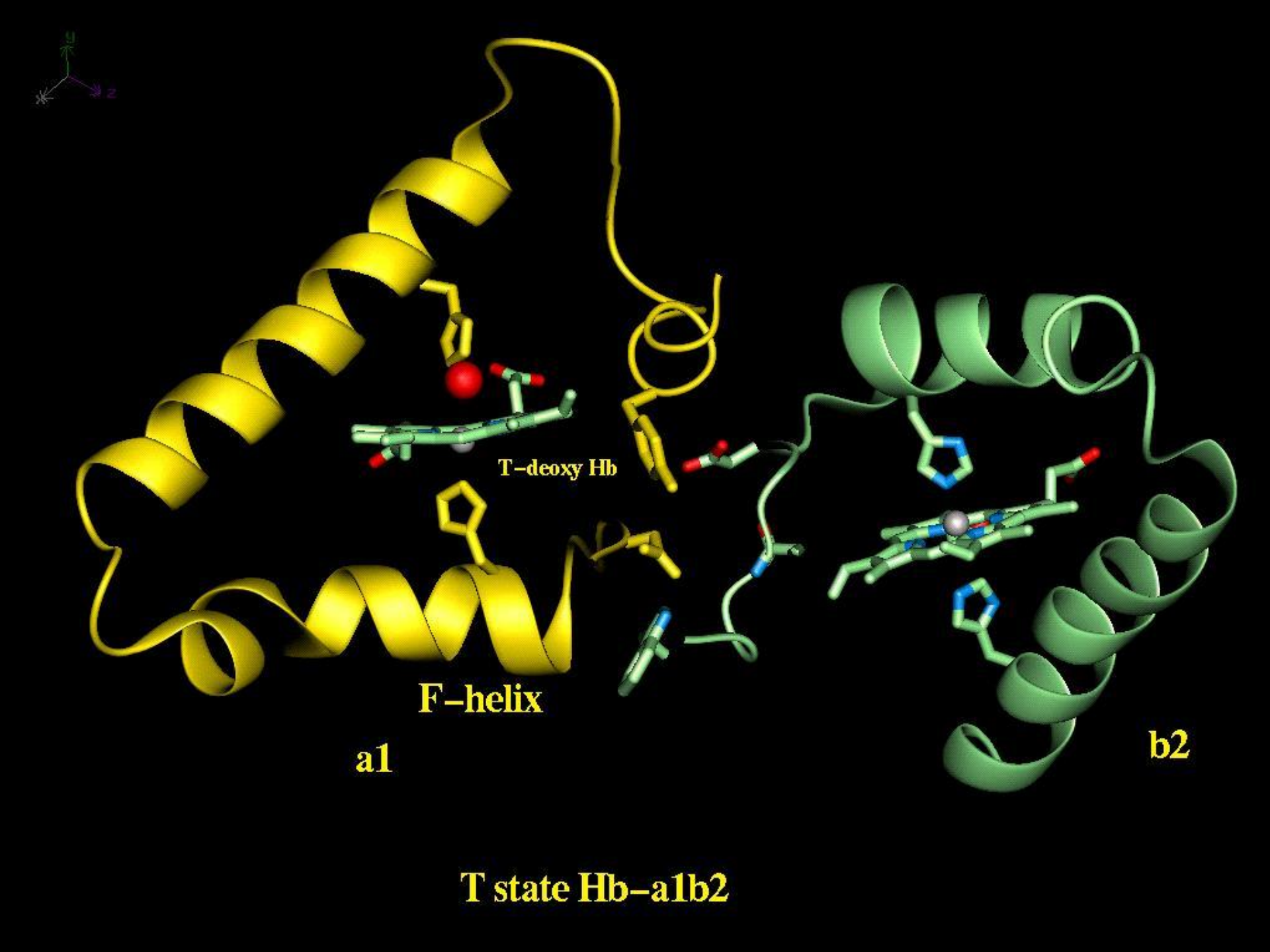
R-state:

Fe(II) d^6 $t_{2g}{}^6$ eg^0 low spin

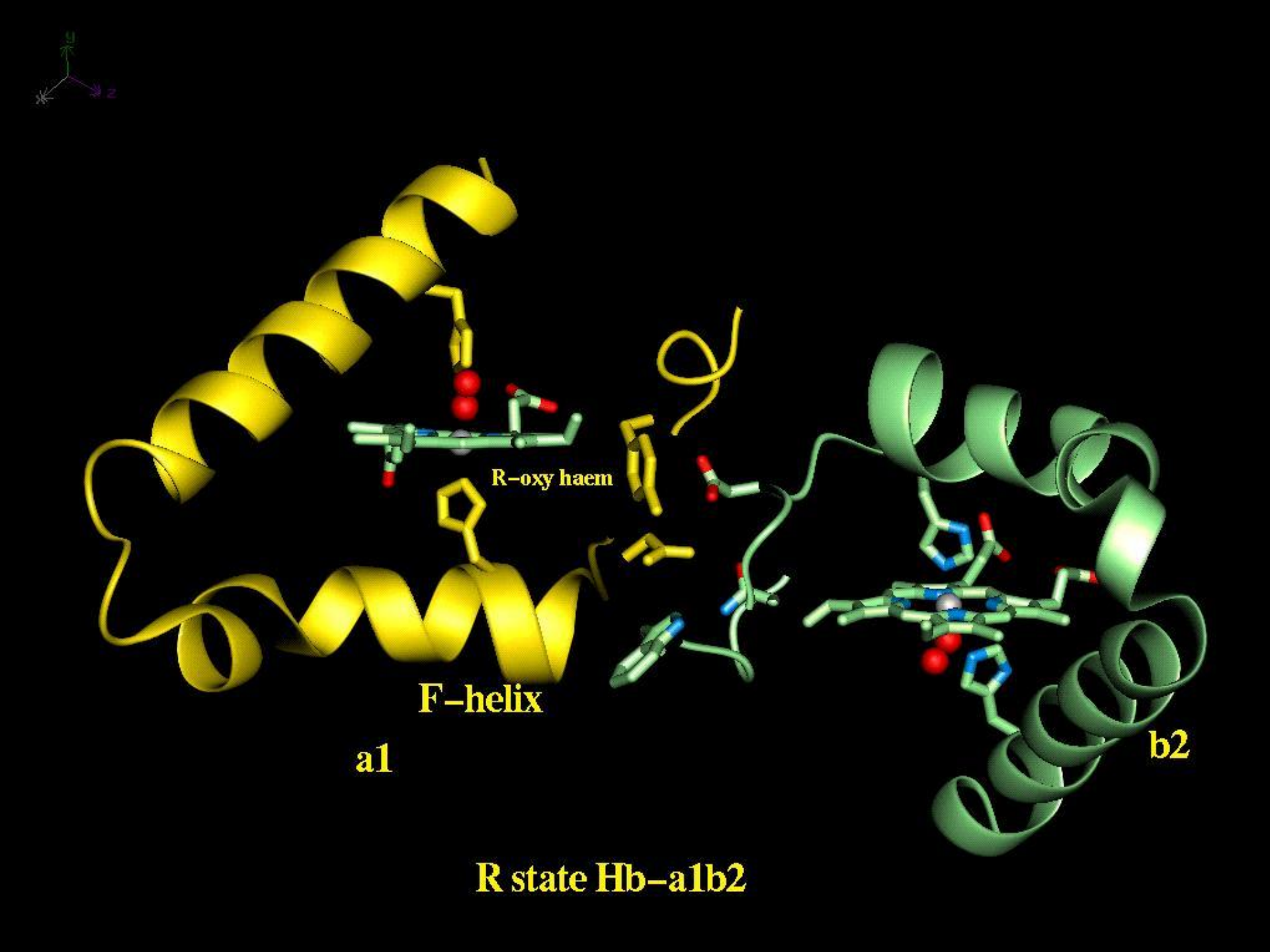
Radius: 75 pm

O2 Binding & Hemoglobin dynamics



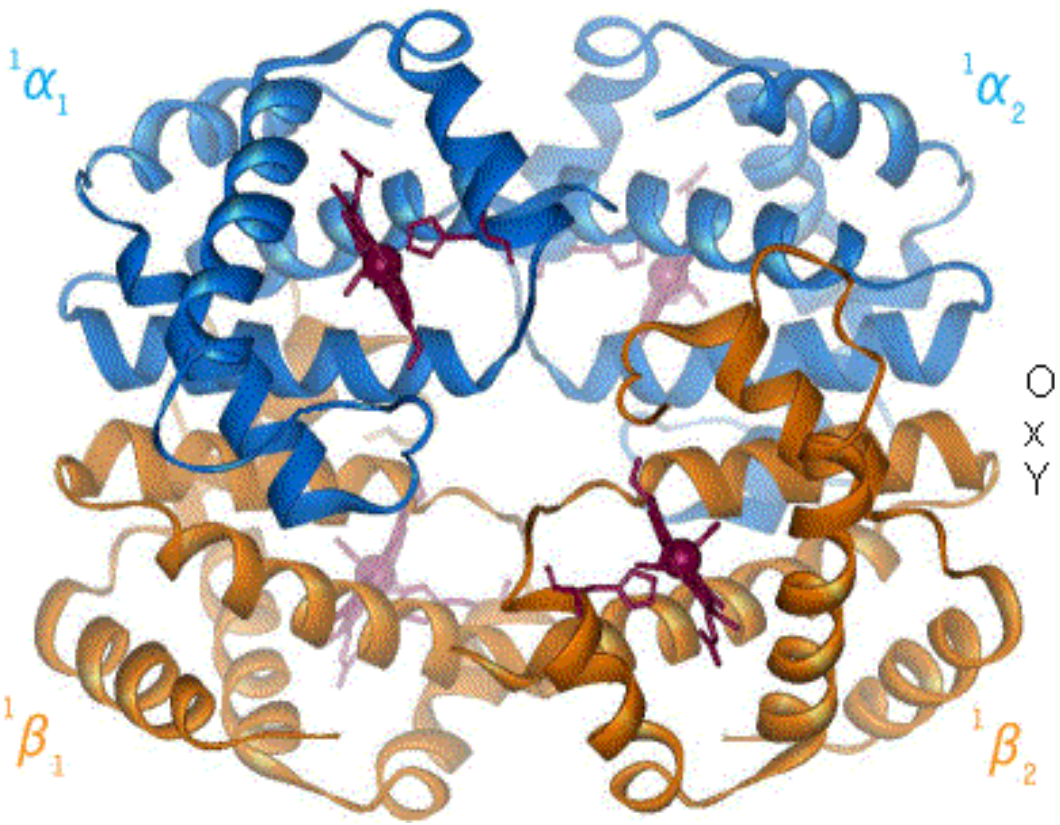


T state Hb-a1b2

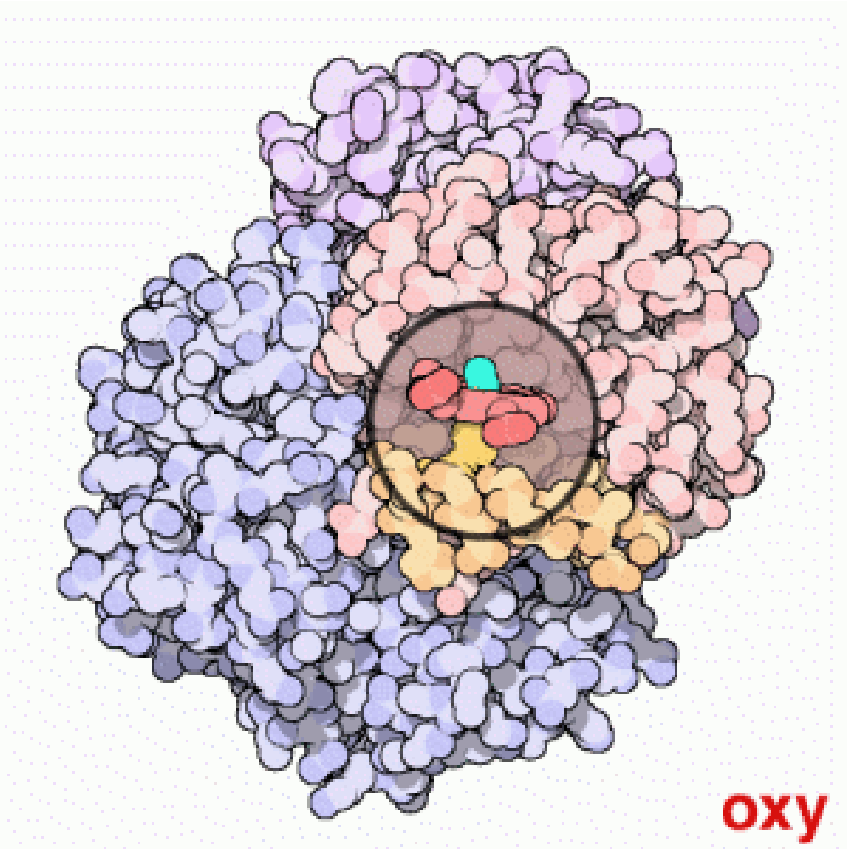


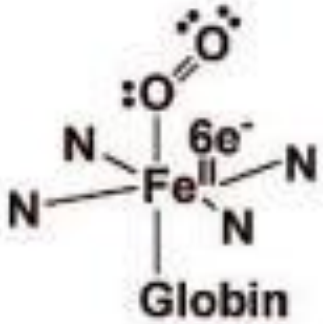
R state Hb–a1b2

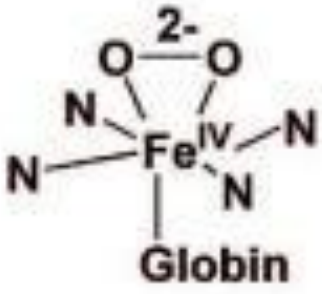
O2 Binding & Hemoglobin dynamics

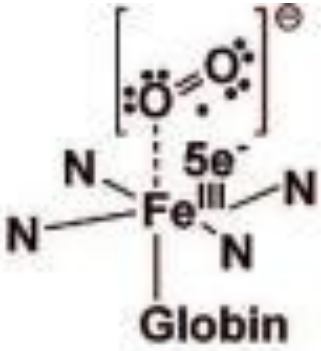


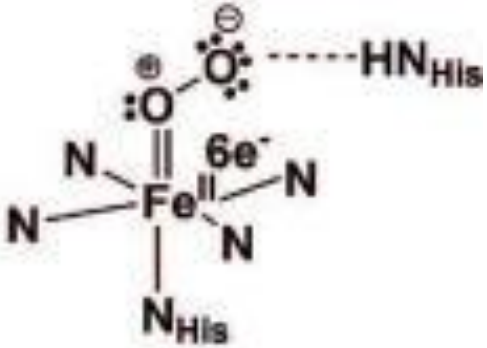
O
X
Y











$Fe^{II}(S=0) \bullet O_2(S=0)$

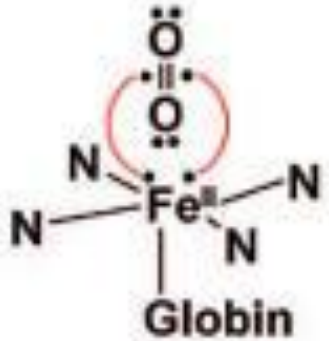
1

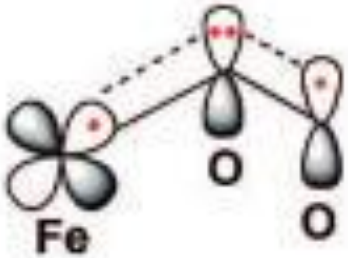
2

$Fe^{III}(S=1/2) \bullet O_2^{-}(S=1/2)$

3

4





$Fe^{II}(S=1) \bullet O_2(S=1)$

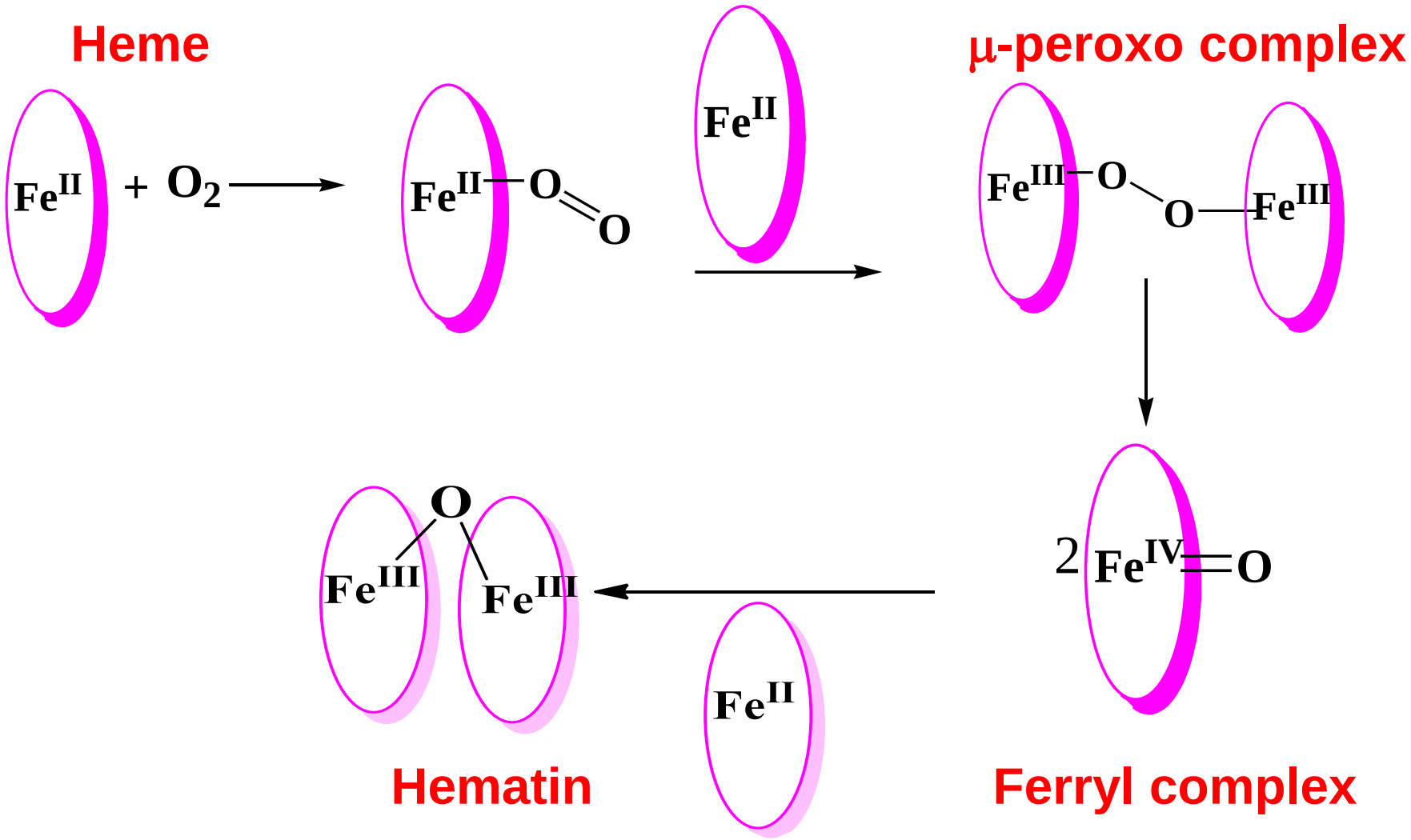
5

$Fe^{II}(S=1) \bullet O_2(S=1)$

6

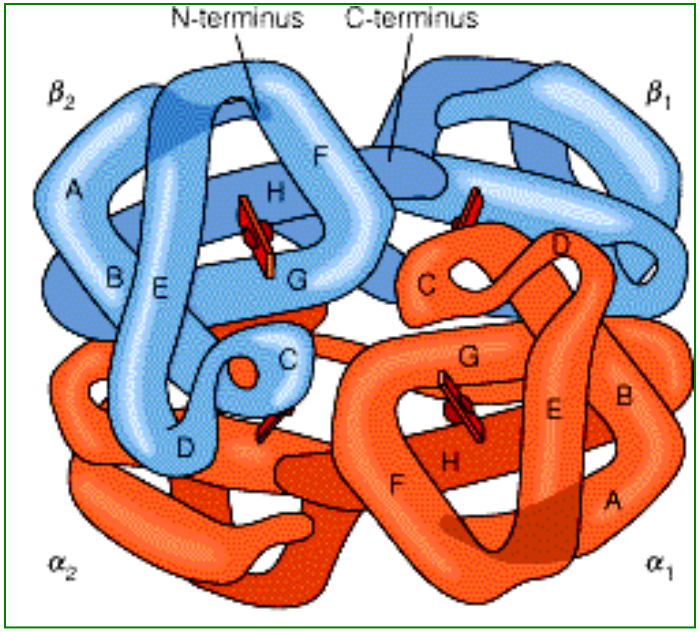
Chen, Shaik et al., J. Am. Chem. Soc. 2008, 130, 14778

O2 Binding & Importance of globin: In a beaker minus globin

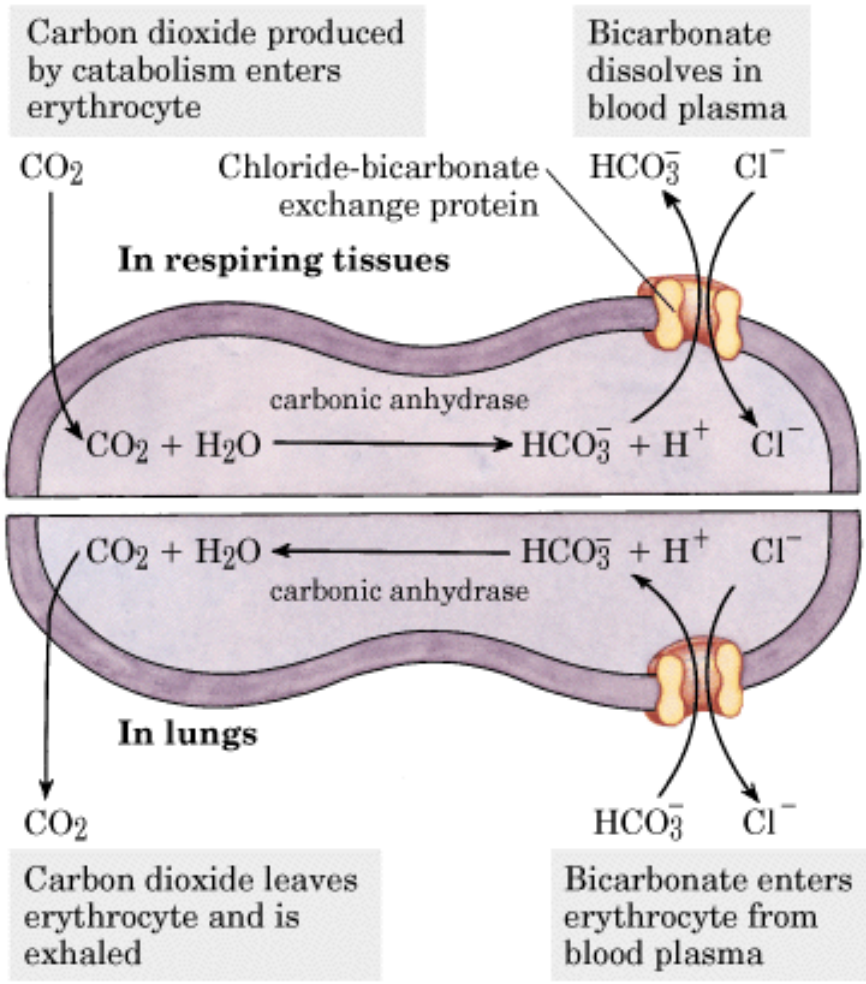


Hemoglobin also carries CO_2 and H^+

- The amino-termini of the four chains become carbamylated in the tissues
- Helping to transport about 20% of the CO_2 dioxide formed in the tissues to the lungs & kidneys
- But what happens to the rest of the CO_2?
- …and how else is the "pH" involved?



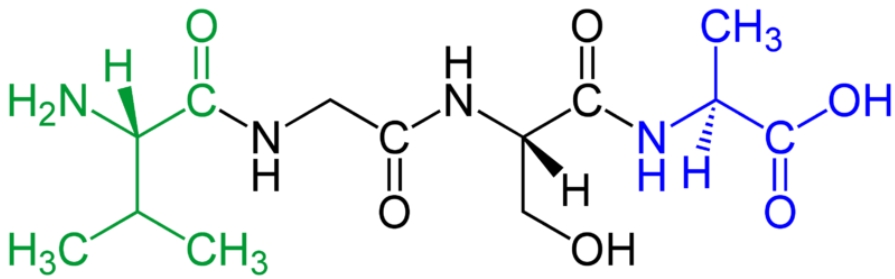
Recall aqueous CO_2 forms carbonic acid



- Carbon dioxide released in tissues is insoluble until hydrated and converted to bicarbonate by enzyme carbonic anhydrase
- But this releases H^+ (causing lowering of pH in the tissues and blood)
- This process is reversed in the lungs …

Bohr effect

an increasing concentration of protons and/or carbon dioxide will reduce the oxygen affinity of hemoglobin

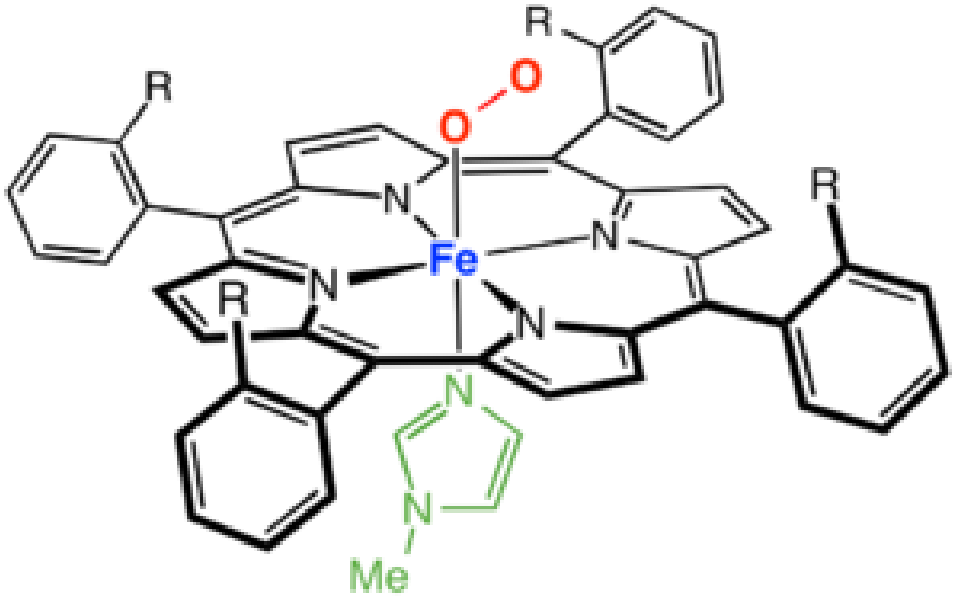


In deoxyhemoglobin, the N-terminal amino groups of the α-subunits and the C-terminal histidine of the β-subunits participate in ion pairs

The formation of ion pairs causes them to decrease in acidity

In oxyhemoglobin, these ion pairings are absent and these groups increase in acidity

Artificial haemoglobin



Terminal Electron Transfer in Respiratory Chain

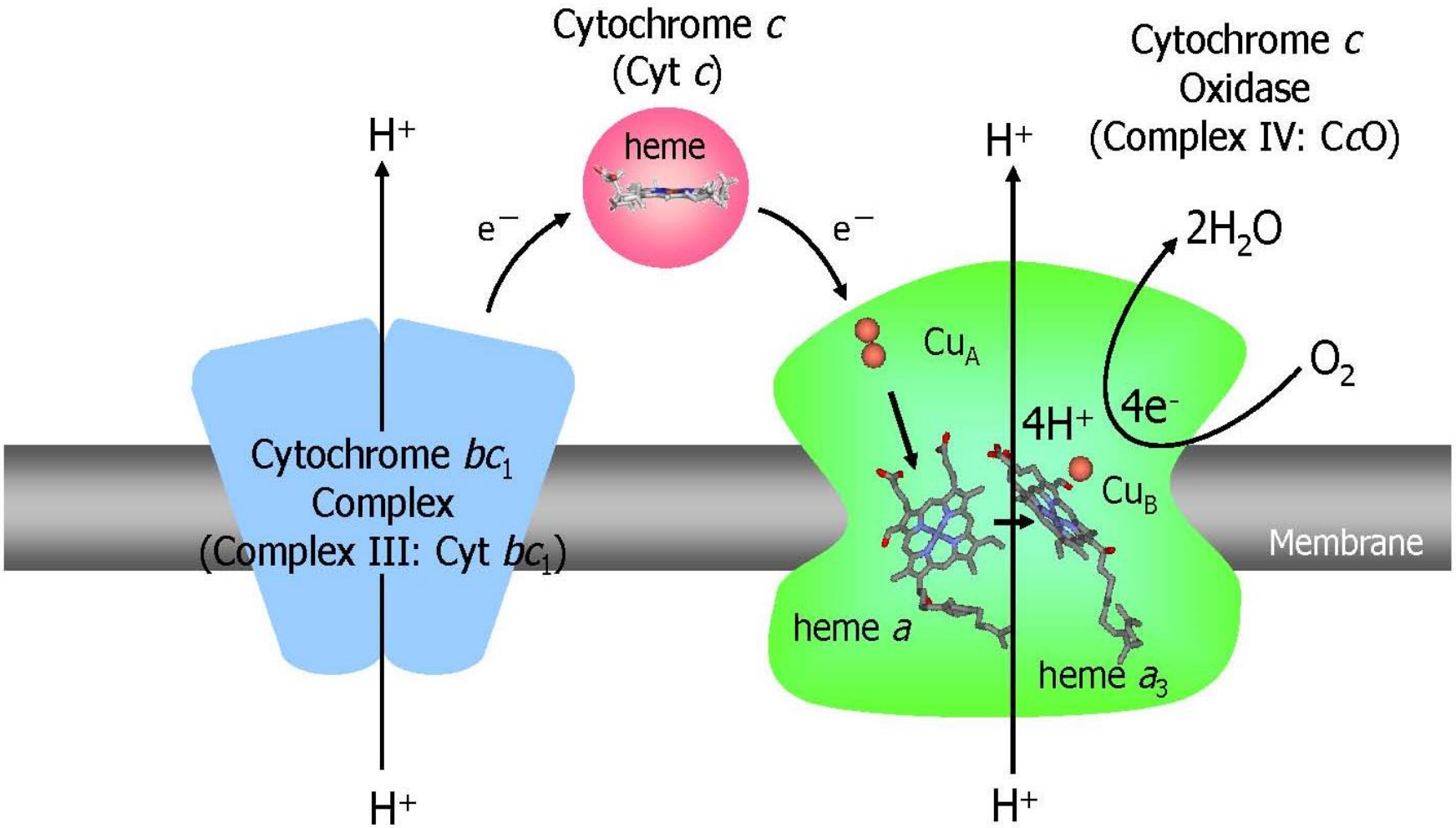
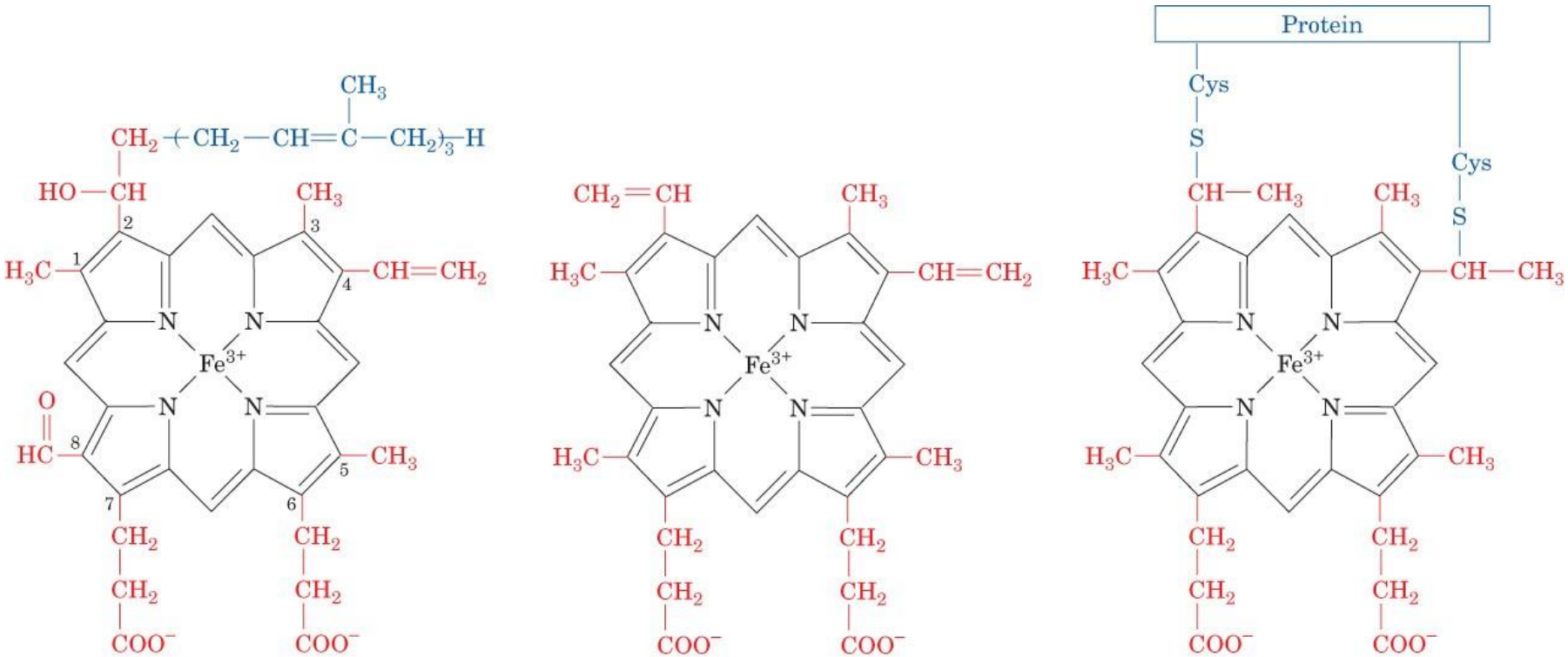
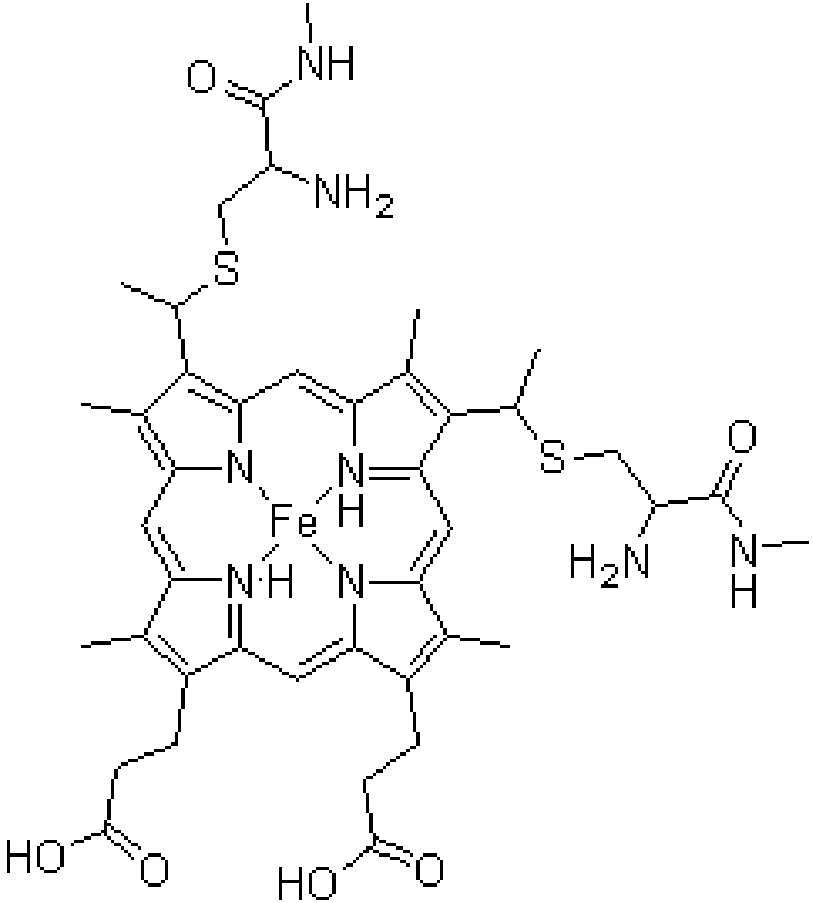
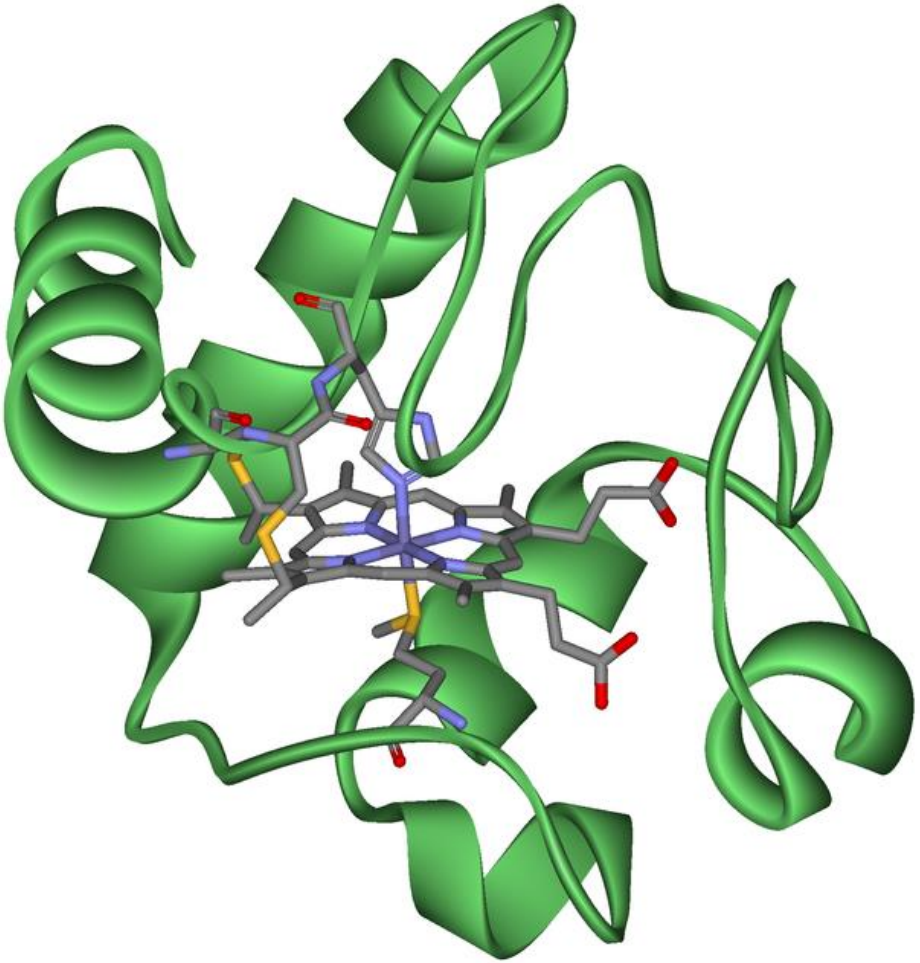


Figure 22-22a Porphyrin rings in cytochromes. (*a*) Chemical structures.



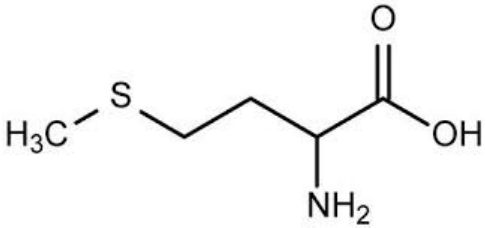


Cytochrome c

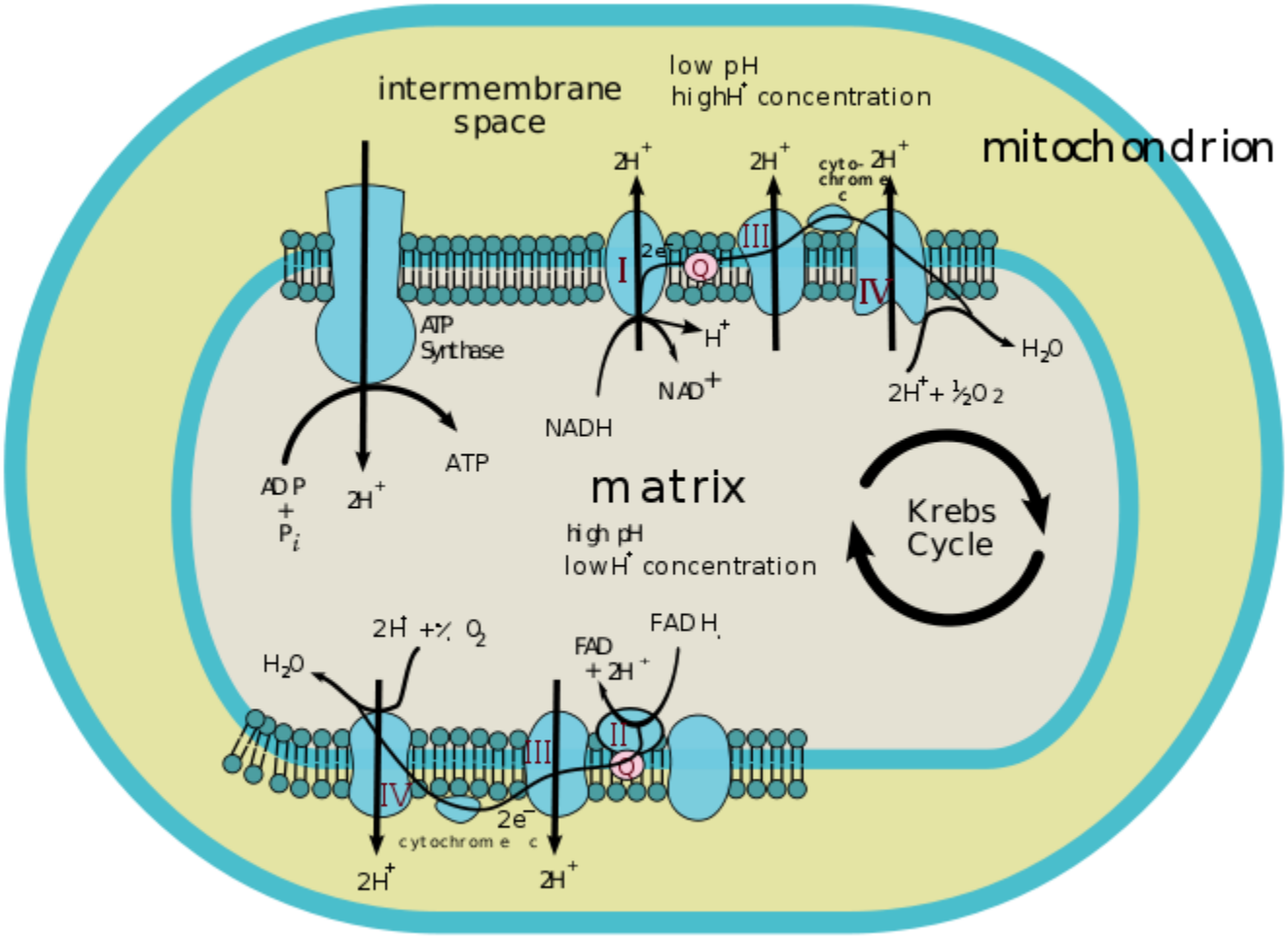


Histidine

Methionene



Mitochondrial Electron Transport Chain

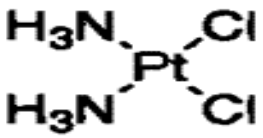


METALS IN MEDICINE

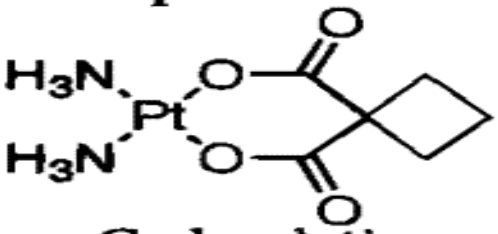
• Li	Maniac depressive illness
• Cu	Anti inflammatory Drugs
• Bi	Peptic Ulcers
• Sn & Zn	Neonatal Jaundice
• V	Diabetics
• Ge(IV), Sb(III)	Hypertension
• As(III)	African Sleeping sickness
• Sb	Anti Parasitic
• Ag	Anti Fungal
• Au	Aurinofin- Arthritis
• Pt	Cisplatin & Carboplatin- Cancer

Chemical Review Vol. 99, Issue 9, 1999

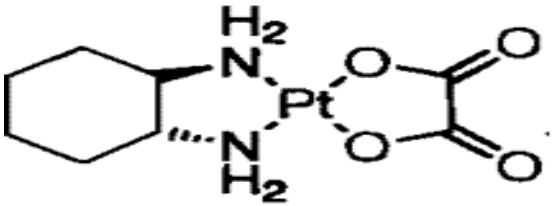
Active Platinum Compounds



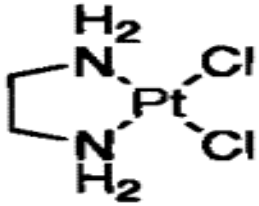
Cisplatin



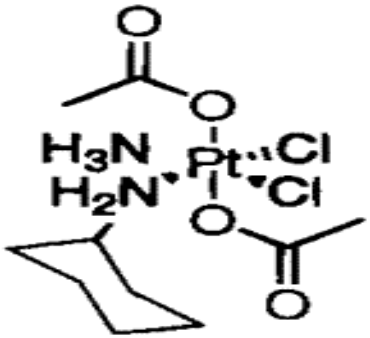
Carboplatin



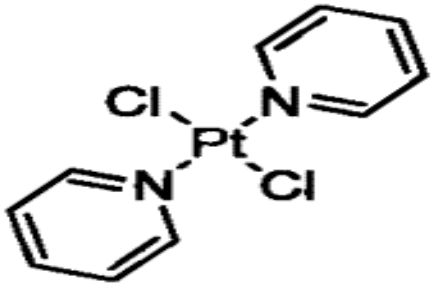
Oxaliplatin



$[Pt(en)Cl_2]$

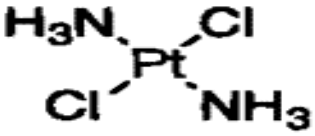


JM 216
Oral Analog

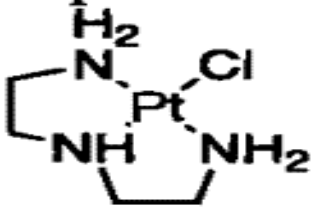


$[Pt(py)_2Cl_2]$
Active trans compound

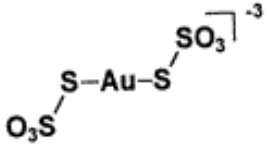
Inactive Platinum Compounds



trans-DDP

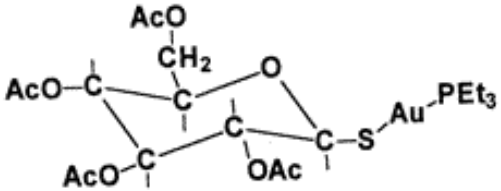


$[Pt(dien)Cl]^+$



bis(thiosulfato)gold(I)

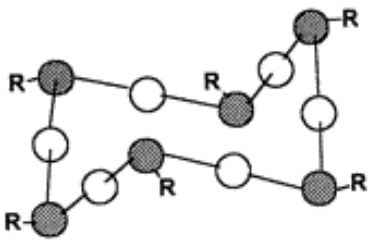
(a)



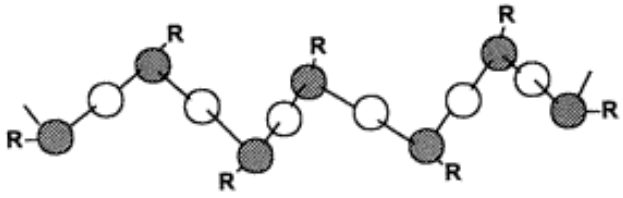
auranofin

(b)

(a-c) antiarthritic drugs
(d) gold(III) antitumor agents
(c) antimalarial



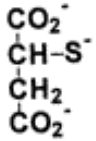
Cyclic Structure



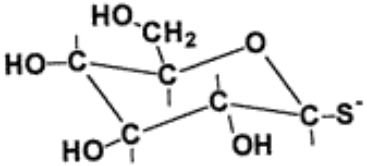
Open Chain

○ = Au ● = S

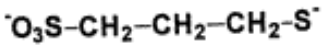
RS⁻ =



Thiomalate H_{-1}

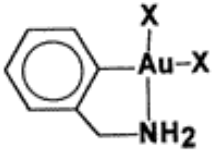


Thioglucose H_{-1}



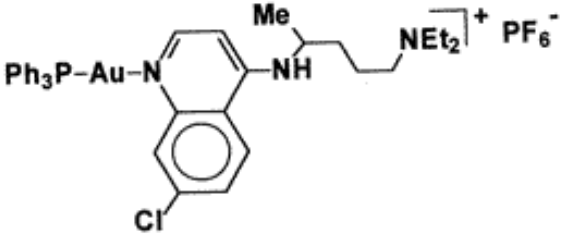
Thiopropanesulfonate H_{-1}

(c)



Au(damp)X_2 (X = Cl^-, OAc^-, etc.)

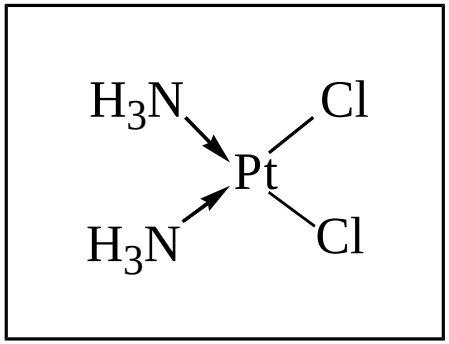
(d)



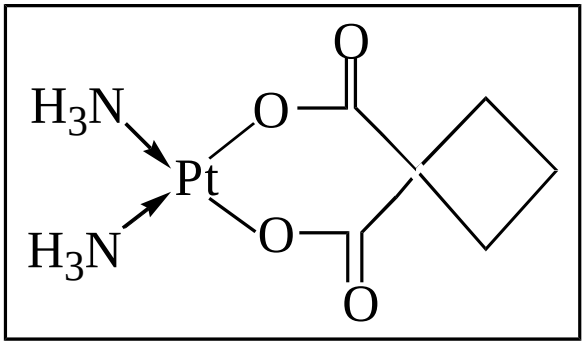
[Ph_3P(chloroquine)gold] PF_6

(e)

Anticancer Pt-Drugs in Chemotherapy



DNA-Binding



CISPLATIN **CARBOPLATIN**

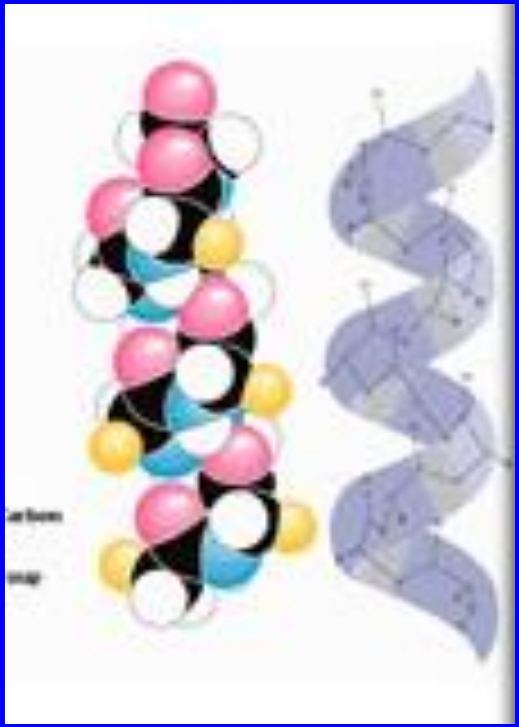
End here

secondary structure proteins

spring

N-H hydrogen bond C=O

amino acid

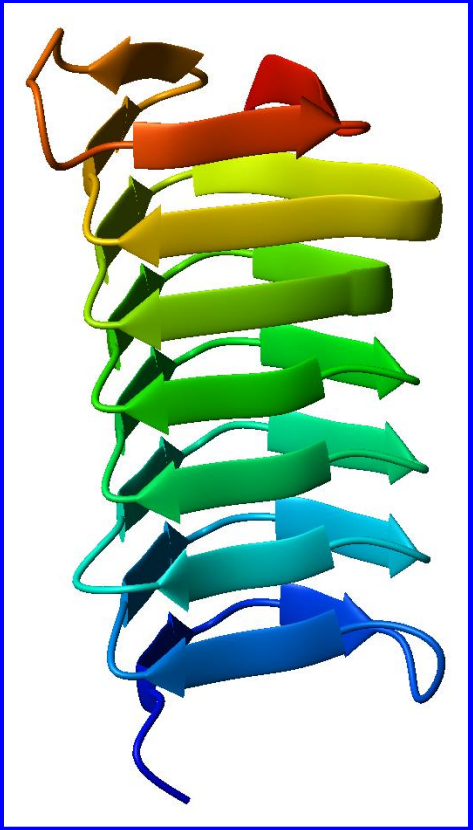


protein

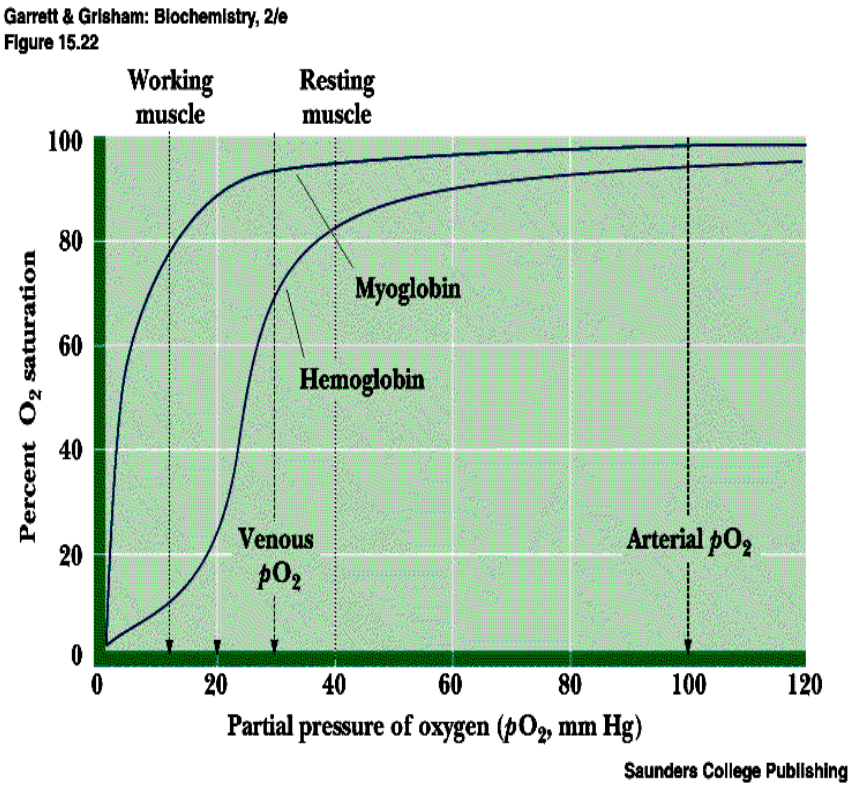
beta strands

hydrogen bonds protein-protein

interactions ions



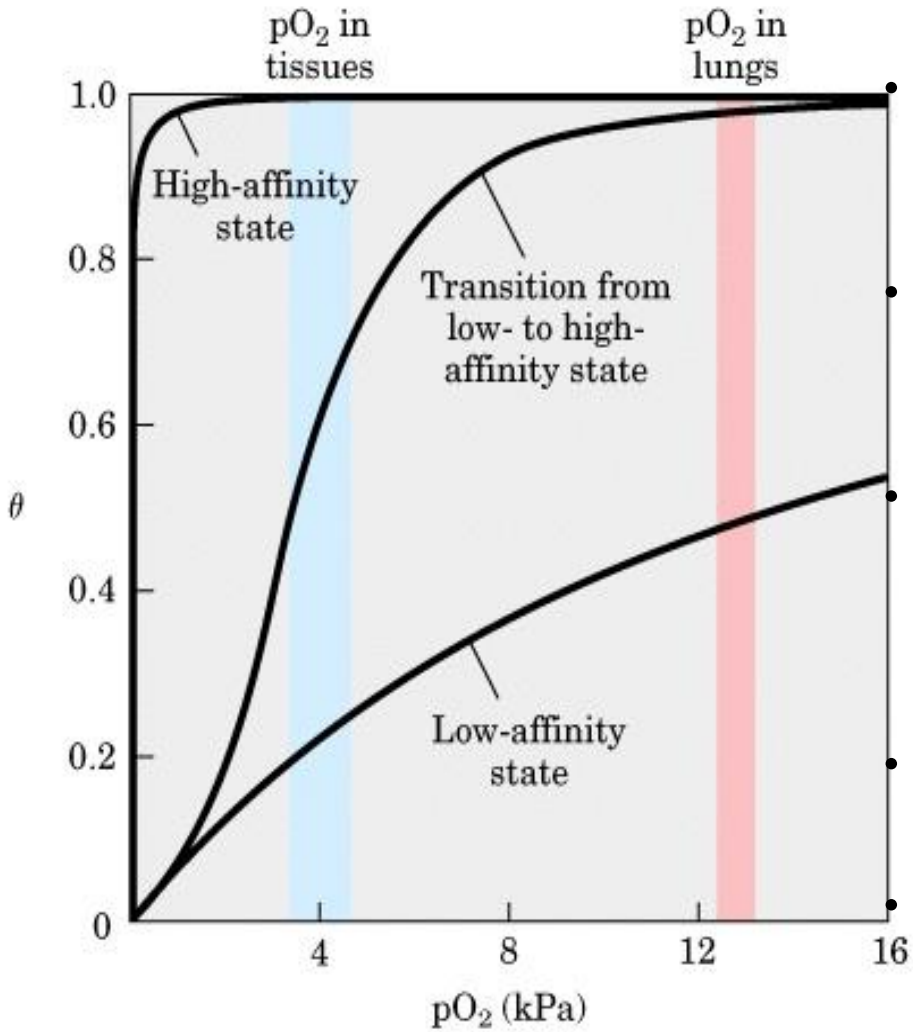
Efficiency in oxygen transport is achieved by cooperative binding in multisite proteins described by a sigmoidal binding curve



$$Hb + 4\,O_2 = Hb(O_2)_4$$

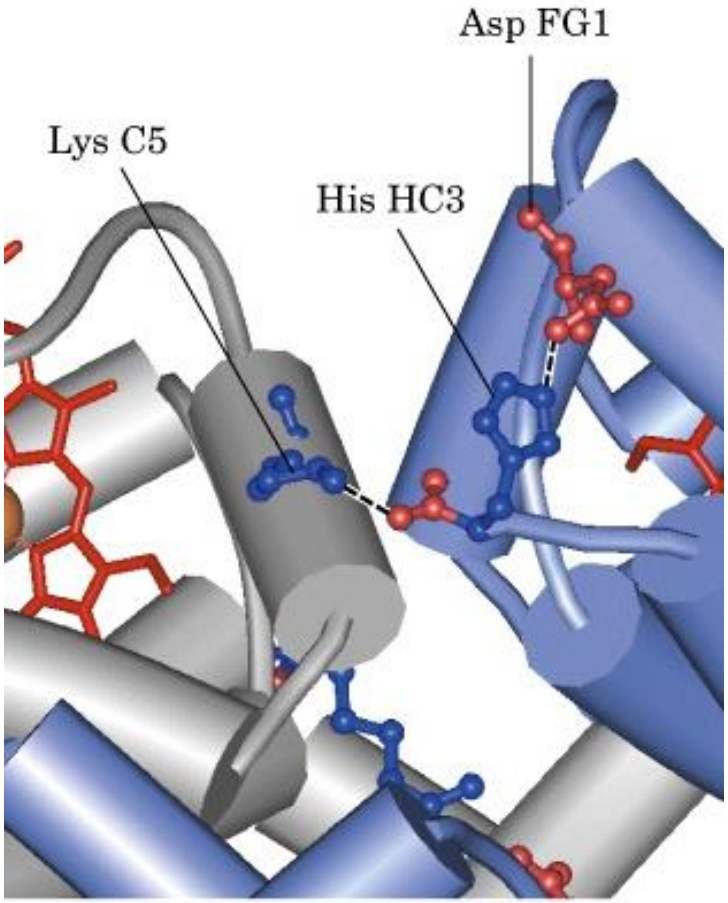
$$K_{Hb} = [Hb(O_2)_4]/[Hb][O_2]^{2.8}$$

- Oxygen binding to hemoglobin is distinctly different from its binding to myoglobin. Notice that the curve is not hyperbolic; it is sigmoidal. This is a sign of cooperativity.
- The job of hemoglobin is to bind oxygen efficiently in the lungs, where the partial pressure of oxyen is about 13.3 kPa; however, it needs to release it in the tissues, where the partial pressure is about 5 kPa.
- Notice how myoglobin is almost fully saturated at 5 kPa pO_2. Therefore, myoglobin would not release its bound oxygen to tissues.
- From looking at the sigmoidal curve for hemoglobin, it can be seen that it binds oxygen relatively weakly at low oxygen concentrations, but strongly at high oxygen concentrations.
- This behavior is referred to as allostery. Hemoglobin is an allosteric protein.



- Deoxyhemoglobin low-affinity T-state picks up O_2 in the lungs, where pO_2 is high,
- Converting it into the high-affinity R-state, so it readily picks up more O_2
- As loaded oxyhemoglobin (R-state) moves back to tissues, where pO_2 is lower, it begins to lose its O_2,
- Which facilitates the transition back to the T-state
- Which facilitates loss of O_2…

Role of Hb in carrying H+

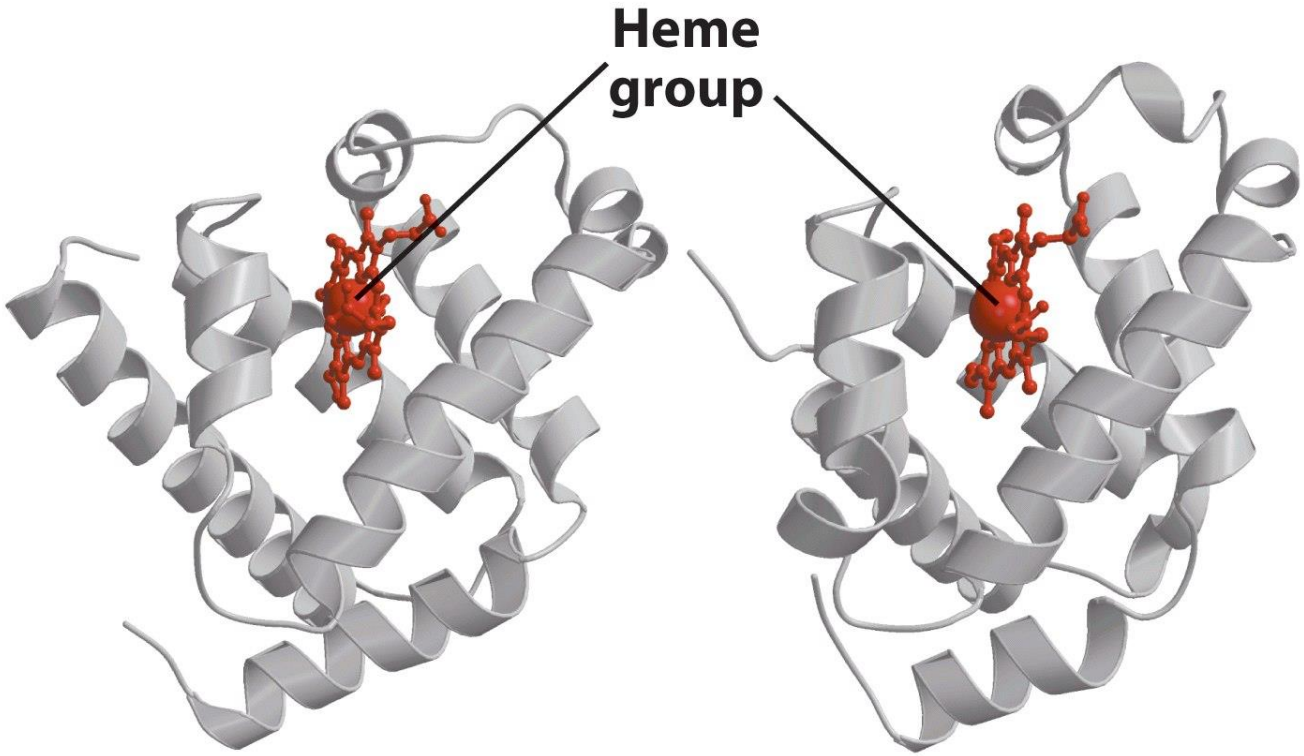


(a)

- Histidine HC3 of T-state hemoglobin has a higher pKa than normal (via interaction with Asp FG1), and thus gets readily protonated
- Back at the lungs, return to the R-state shifts HC3 back to its normal pKa, releasing the H^+
- Release of H^+ promotes conversion to the R state, increasing O_2 affinity…

Hemoglobin structure

Myoglobin is a monomer. Each hemoglobin protomer resembles myoglobin:



Inorganic Chemistry & Environment

Acid Rain: Tajmahal is in danger

- combustion of high sulfur fuels, roasting of sulfur ores like Cu_2S

  $\Rightarrow$ Sulfur oxides $\Rightarrow$ sulfuric acid

- Auto Exhaust $\Rightarrow$ N_2+O_2 $\Rightarrow$ Oxides of Nitrogen NOx $\Rightarrow$ nitric acid

- Dissolved acids in Environment $\Rightarrow$ "acid rain"

- Some times, the pH is as low as 1.5

Inorganic Chemistry & Environment

Depletion of Ozone Layer:

- **Nature ⇒ Photochemical process**

$$\Downarrow$$

Formation and Destruction of ozone

$$O_2 + h\nu \text{ (below 242 nm)} \rightarrow O\cdot + O\cdot$$

$$O\cdot + O_2 \rightarrow O_3$$

$$O_3 + h\nu \text{ (220 - 320 nm)} \rightarrow O\cdot + O_2$$

- **CULPRIT: NO & CFC (F_3CCl)**

Inorganic Chemistry & Environment

Depletion of Ozone Layer:

Supersonic Concorde

$\Downarrow$

NO

$$NO + O_3 \rightarrow NO_2 + 2O^{\cdot}$$

$$NO_2 + O^{\cdot} \rightarrow NO + O_2$$

Refrigerant

$\Downarrow$

CFC

$$F_3CCl \rightarrow F_3C^{\cdot} + Cl^{\cdot}$$

$$Cl^{\cdot} + O_3 \rightarrow ClO^{\cdot} + O_2$$

$$ClO^{\cdot} + O^{\cdot} \rightarrow Cl^{\cdot} + O_2$$

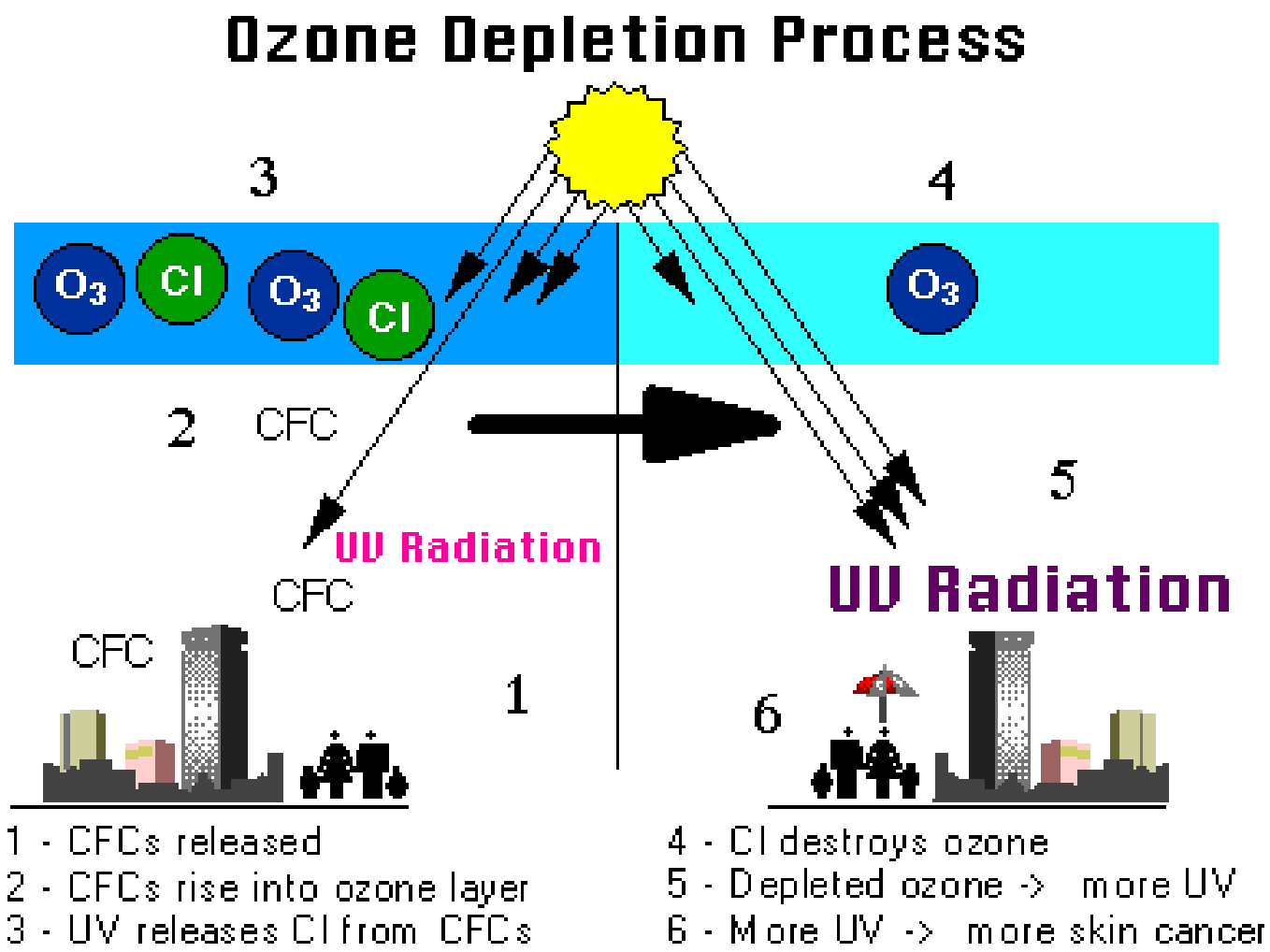
The net reaction is

$$O^{\cdot} + O_3 \rightarrow 2O_2$$

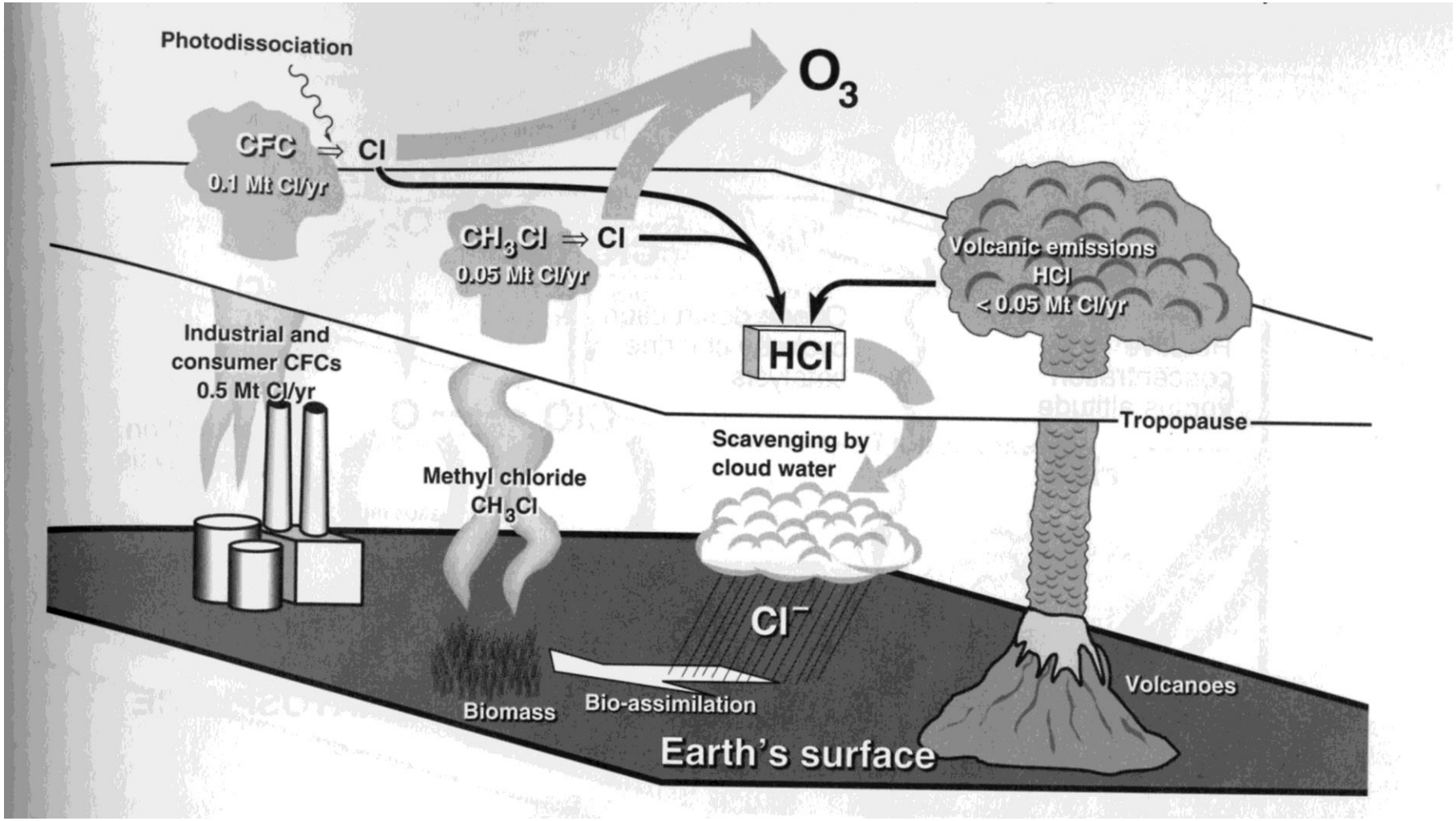
What causes the depletion?

- release of manmade chemicals –
- **CFC - refrigerants, aerosol sprays, solvents and foam-blowing agents**
- halogen compounds – **Fire extinguisher**
- **Halon extinguishers normally contain bromochlorodifluoromethane**
- NO_x

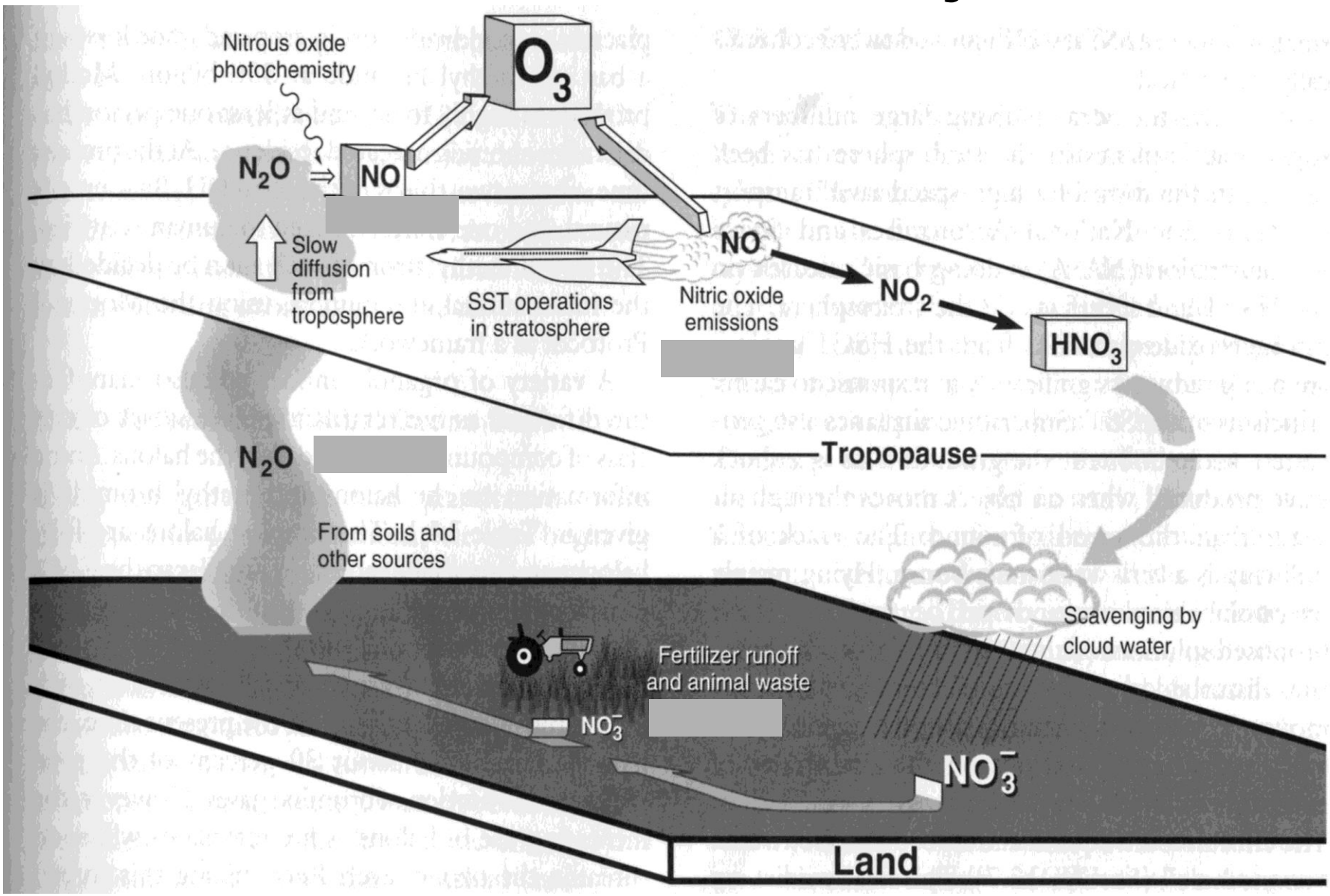
One chlorine atom - average - destroys one thousand ozone molecules



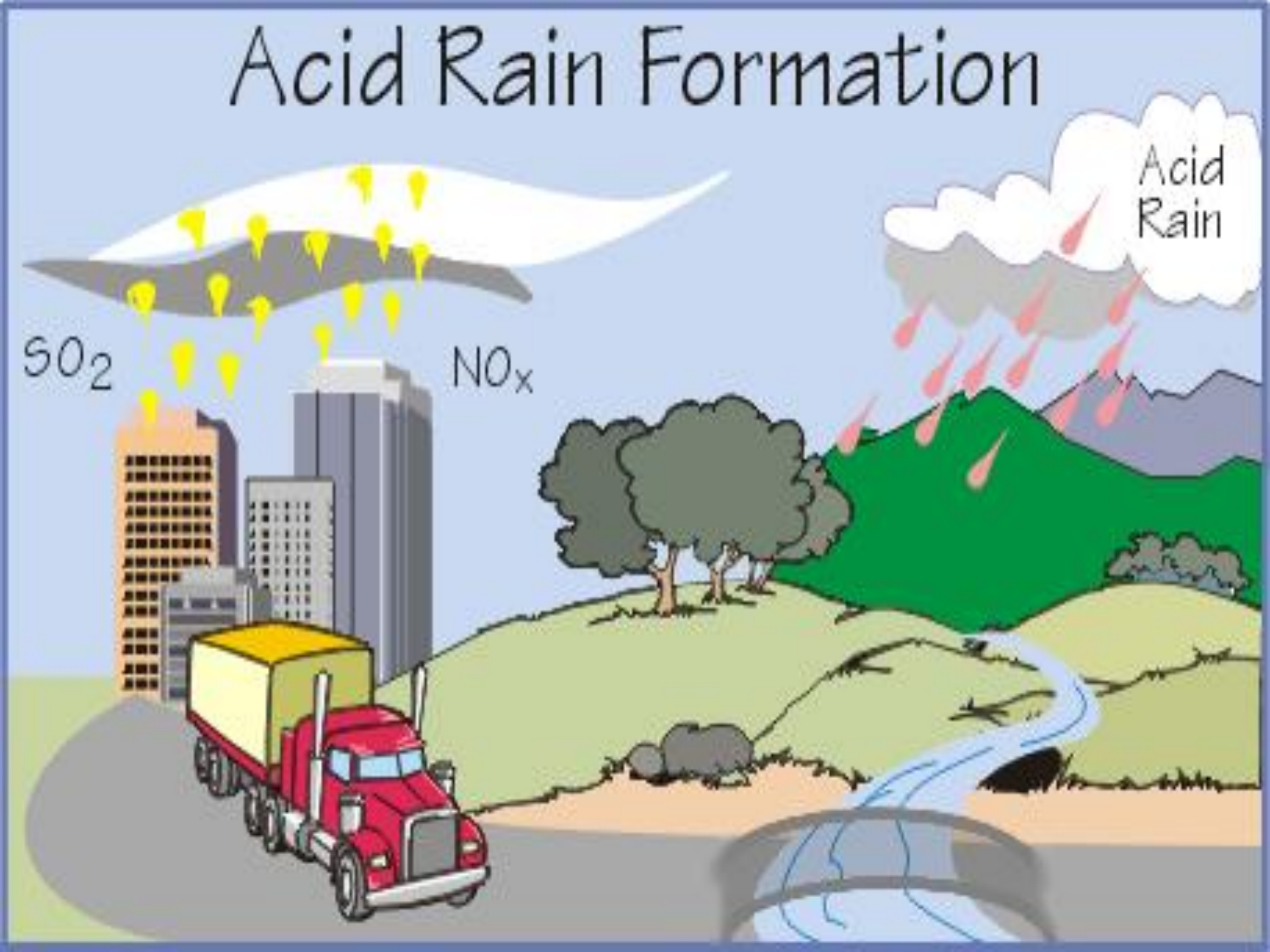
Atmospheric Chlorine cycle

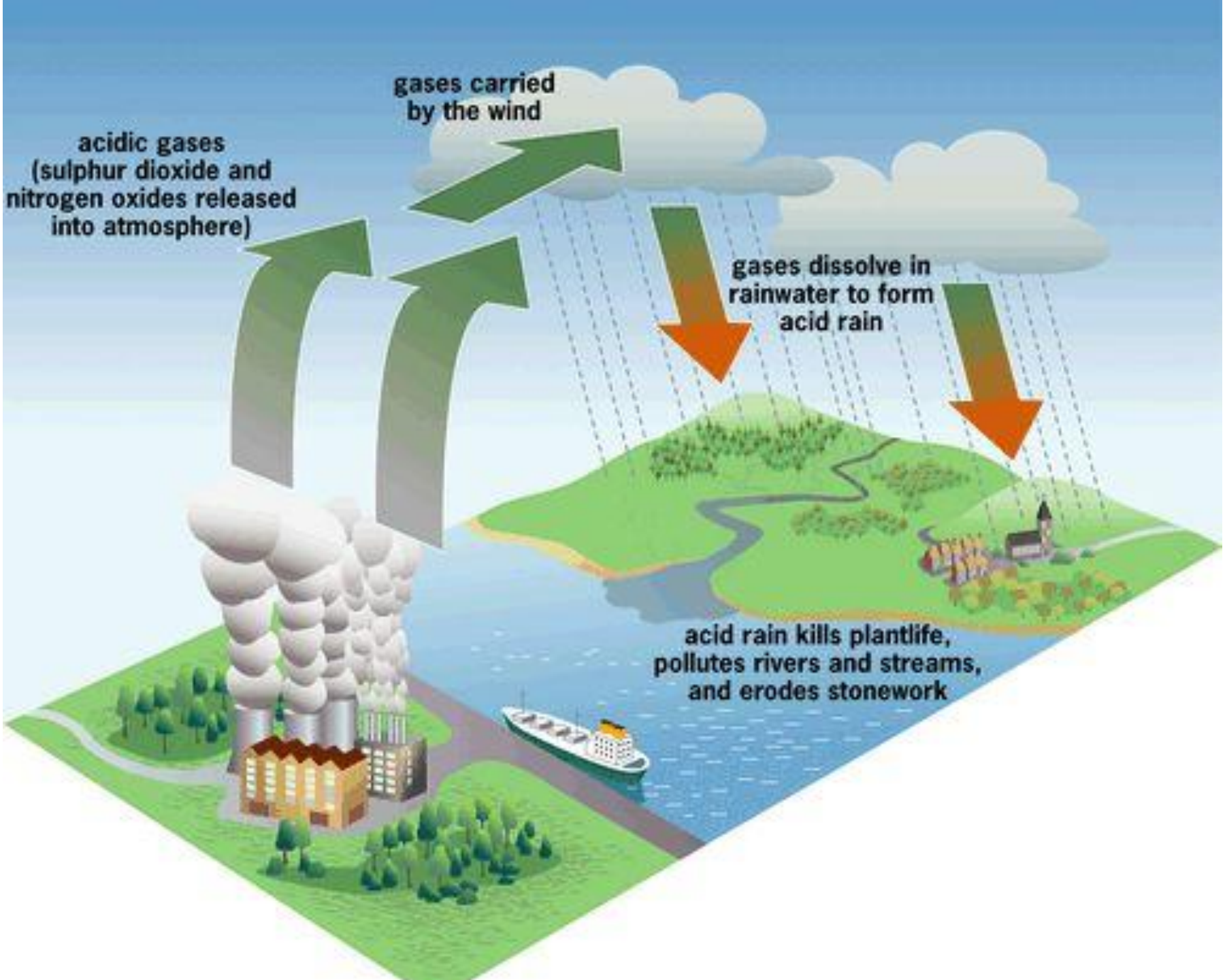


Nitrous oxide cycle



Acid Rain Formation




gases carried
by the wind
acidic gases
(sulphur dioxide and
nitrogen oxides released
into atmosphere)
gases dissolve in
rainwater to form
acid rain
acid rain kills plantlife,
pollutes rivers and streams,
and erodes stonework